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ORIGINAL PAPER

Zeta zeros and prolate wave operators

Semilocal adelic operators

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Abstract

We integrate in the framework of the semilocal trace formula two recent discoveries on the spectral realization of the zeros of the Riemann zeta function by introducing a semilocal analogue of the prolate wave operator. The latter plays a key role both in the spectral realization of the low lying zeros of zeta—using the positive part of its spectrum—and of their ultraviolet behavior—using the Sonin space which corresponds to the negative part of the spectrum. In the archimedean case the prolate operator is the sum of the square of the scaling operator with the grading of orthogonal polynomials, and we show that this formulation extends to the semilocal case. We prove the stability of the semilocal Sonin space under the increase of the finite set of places which govern the semilocal framework and describe their relation with Hilbert spaces of entire functions. Finally, we relate the prolate operator to the metaplectic representation of the double cover of $\mathrm{SL}(2, \mathbb{R})$ with the goal of obtaining (in a forthcoming paper) a second candidate for the semilocal prolate operator.

Keywords Riemann zeta · Semilocal trace formula · Sonin space · Prolate operator · Orthogonal polynomials · Metaplectic representation · Jacobi matrix · Infrared · Ultraviolet

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Contents

1 Introduction

The difficulty of solving the Riemann Hypothesis (RH) is often primarily attributed to the infinite number of terms within the Euler product.

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1} \tag{1.1}$$

However, contrary to this widespread belief, there exists a property $P(n)$, involving only the Euler factors for primes smaller than n, and whose validity for all n is equivalent to RH. This is derived from Weil's positivity criterion (see [20, 23]) which involves the quadratic form Q_n defined using the Riemann-Weil explicit formulas applied to test functions φ with support in the compact symmetric interval $[n^{-1/2}, n^{1/2}]$. Indeed, by construction the evaluation of the explicit formula on the function $\varphi * \varphi^*$ only involves the primes $p \leq n$ and one can take for $P(n)$ the property

$$Q_n(\varphi, \varphi) \geq 0, \ \forall \varphi \in C_c^\infty(\mathbb{R}_+^*) \mid \text{Support}(\varphi) \subset [n^{-1/2}, n^{1/2}]. \tag{1.2}$$

Furthermore, the semilocal[1] trace formula of [5] gives, for each n, a Hilbert space theoretic framework in which the Weil quadratic form Q_n becomes the trace of a simple operator theoretic expression, thus offering a natural ground of exploration in order to attack RH.

The present paper advances the operator-theoretic approach to RH by proposing a unified framework that integrates two recent breakthroughs on the spectral realization of the zeros of the Riemann zeta function, which we will now review.

On the one hand, the results of [7, 8] have shown that the semilocal trace formula of [5] gives a conceptual explanation of Weil's positivity for the archimedean place, the source of it being the Hilbert space representation of the scaling group. Moreover, in the same vein, the results in [9] showed that one can access to the infrared (low lying) part of the zeros of the Riemann zeta function using the scaling operator with periodic boundary conditions, restricted to the orthogonal of the radical of the Weil quadratic form. On the other hand, the unusual ultraviolet behavior of the zeros of the Riemann zeta function has found an unexpected spectral incarnation in [10] in

[1] The term semilocal is used throughout the text when a finite set S of places of $\mathbb{Q}$ containing the archimedean place, is involved, as well as the ring which is the product of the local fields $\mathbb{Q}_v$, $v \in S$.

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

terms of the *negative* eigenvalues of the classical prolate wave operator extended to the whole real line.

The goal of this paper is to integrate these two developments and to extend our analysis beyond the single archimedean place by employing the semilocal framework of [7]. This framework provides a canonical Hilbert space operator setting where all the main concepts remain coherent. Additionally, it offers a conceptual interpretation for both the explicit formula and the radical of the Weil quadratic form.

At the forefront of this strategy stands the pivotal role of the prolate operator which we already proved to be crucial in both (infrared and ultraviolet) observed agreements with the zeros of zeta. Its significance in [9] stemmed from the fact that the radical of the Weil quadratic form is determined by the range of the map $\mathcal{E}$. This map is defined on the even Schwartz space $\mathcal{S}(\mathbb{R})_0^{\rm ev}$ by the formula $\mathcal{E}(f)(u) := u^{1/2}\sum f(nu)$. When $\mathcal{E}$ is applied to the eigenfunctions φ_n of the prolate operator, associated with *positive* eigenvalues, it results in exceptionally small eigenvalues of the Weil quadratic form restricted to functions that have specified compact support. The corresponding eigenfunctions ψ_n of the Weil quadratic form were then reconstructed in [9] through the Gram–Schmidt orthogonalization process applied to $\mathcal{E}(\varphi_n)$. Incidentally, the semilocal form of the map $\mathcal{E}$ agrees with the canonical identification of the semilocal functions with functions on idele classes. The conceptual meaning of $\mathcal{E}$ becomes clear in the semilocal formalism. The aforementioned eigenfunctions ψ_n were used in [9] to condition the scaling operator with periodic boundary conditions. With this process we generated the low-lying zeros of the zeta function, while the ultraviolet behavior of the zeros remained out of reach. Since the expected UV behavior has been realized by the negative spectrum of the prolate operator, the next challenge is now to find the semilocal analogue, at least at the formal level, of that operator. We expect that the use of such operator-theoretic tools in the semilocal case opens a way to handle Weil's positivity as in [8]. In fact, the operator theoretic aspect of the present paper provides a more precise strategy for addressing the semilocal Weil positivity by comparing the trace functional associated to the operator—which is automatically positive for a selfadjoint operator—with the Weil functional. The conditioning, by the radical of the Weil quadratic form, that worked for the scaling operator in the infrared case will be implemented automatically by the orthogonality of the positive and the negative part of the spectrum of the semilocal prolate operator, whose corresponding negative eigenspace[2] was identified in [10] to the Sonin space.

In summary, our strategy aims to reinterpret the prolate operator in order to suggest a semilocal equivalent. This is achieved by prioritizing the roles of the scaling operator, whose semilocal version is well established in [5], and that of the Sonin space.

The scaling operator $\mathbb{S}$ is the selfadjoint generator of the group of dilations in the Hilbert space $L^2(\mathbb{R})^{ev}$ of square integrable even functions on the real line. The prolate operator admits a simple expression in terms of the operator $\mathbb{S}$ and the vector given by the function $\xi(x) := e^{-\pi x^2}$. This vector belongs to the domain of $\mathbb{S}^n$ for any n, and the span of the $\mathbb{S}^n\xi$ is $L^2(\mathbb{R})^{ev}$. We let N be the grading associated to the filtration coming from the linear span of polynomials $p(\mathbb{S})\xi$, with deg $P \le n$. Ignoring the delicate domain definition needed to obtain a selfadjoint operator, the prolate operator

[2] Up to a finite dimensional possible discrepancy.

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

$\mathbf{W}_\lambda$ is given by the formal expression

$$\mathbf{W}_\lambda = -\mathbb{S}^2 + 2\pi\lambda^2(4N+1) - \frac{1}{4}. \tag{1.3}$$

Formula (1.3) is meaningful for any *cyclic pair* (D, ξ) given by a selfadjoint operator D acting in a Hilbert space $\mathcal{H}$ and a unit vector $\xi \in \cap_{n\in\mathbb{N}}\mathrm{Dom}D^n$ which is cyclic (i.e. such that the vectors $p(D)\xi$ with p a polynomial, are dense in $\mathcal{H}$). For any cyclic pair (D, ξ), the spectral theorem provides a *canonical form* (or diagonalization) in which D is represented as the multiplication by the variable $s \in \mathbb{R}$ in the Hilbert space $\mathcal{H} = L^2(\mathbb{R}, d\mu)$ where $d\mu$ is the spectral measure defined by $\int f(s)d\mu(s) := \langle f(D)\xi|\xi\rangle$, and the cyclic vector ξ is the constant function 1. The theory of orthogonal polynomials for the measure $d\mu$ then provides the representation of D as a Jacobi matrix and of the grading operator N as a diagonal matrix. We recall these basic facts in Sect. 2.

It turns out that one can naturally associate to a cyclic pair a spectral triple $(\mathcal{A}, \mathcal{H}, D)$ where $\mathcal{H}$ and D are unchanged while the algebra $\mathcal{A} := c_0(\mathbb{N})$ acts via the grading operator N associated to the orthogonal polynomials of the spectral measure. The role of the spectral triple $(\mathcal{A}, \mathcal{H}, D)$ is to understand algebraically the prolate operator as a perturbation of the operator $-D^2$ by the addition of an element of the algebra $\mathcal{A}$. The spectral triple $(\mathcal{A}, \mathcal{H}, D)$ of a cyclic pair is *even* (in the sense that it admits a $\mathbb{Z}/2\mathbb{Z}$-grading commuting with the algebra and anticommuting with D) if and only if the spectral measure $d\mu$ is even, i.e. invariant under $s \mapsto -s$. Moreover such a $\mathbb{Z}/2\mathbb{Z}$-grading γ is unique if it exists. In the even case one obtains a Jacobi matrix representation for the prolate operator (Proposition 2.4).

In Sect. 3 we describe in Theorem 3.1 the canonical form of the even cyclic pair associated to the scaling operator $D = \mathbb{S}$ and the cyclic vector $\xi_\infty(x) = 2^{1/4}e^{-\pi x^2}$. The grading is given by the Fourier transform $\mathbb{F}_{e_\mathbb{R}}$ acting in $L^2(\mathbb{R})^{ev}$. The isomorphism with the canonical form is given by the unitary

$$\mathcal{V} = \mathcal{M} \circ \mathcal{U} : L^2(\mathbb{R})^{ev} \to L^2(\mathbb{R}, \mathrm{d}m) \tag{1.4}$$

which is the composition of the unitary transformation $\mathcal{U} : L^2(\mathbb{R})^{ev} \to L^2(\mathbb{R})$

$$\mathcal{U}(f) = \pi^{-\frac{1}{2}} \int_0^\infty f(v)v^{\frac{1}{2}-is}\mathrm{d}^*v.$$

with the multiplication operator $\mathcal{M}(f)(s) := \mathcal{U}(\xi_\infty)^{-1}(s)f(s)$ which is a unitary isomorphism $\mathcal{M} : L^2(\mathbb{R}) \to L^2(\mathbb{R}, \mathrm{d}m)$ for the probability measure $\mathrm{d}m(s) := (2\pi)^{-\frac{3}{2}}|\Gamma(\frac{1}{4} + \frac{is}{2})|^2\mathrm{d}s$ on $\mathbb{R}$.

The unitary isomorphism $\mathcal{V}$ of (1.4) is the Hardy–Titchmarsh transform which, modulo normalizations, appears already in the paper [14, Theorem 1, p. 201]. Their original motivation was to construct self-reciprocal functions by conjugating the Fourier transform with the symmetry $s \mapsto -s$. This technique puts in evidence the role of the evenness, i.e. the $\mathbb{Z}/2\mathbb{Z}$-grading, of the cyclic pair. The unitary isomorphism $\mathcal{V}$ transforms the Hermite functions into the orthogonal polynomials P_n for the measure

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

dm and the scaling operator $\mathbb{S}$ into the multiplication by s. Moreover the prolate wave operator

$$(\mathbf{W}_\lambda \psi)(q) = -\partial((\lambda^2 - q^2)\partial)\,\psi(q) + (2\pi\lambda q)^2\,\psi(q) \tag{1.5}$$

becomes a special case of (1.3) and, in fact, of a general construction for orthogonal polynomials. More precisely, with N the grading operator $N(P_n) := n\ P_n$ in $L^2(\mathbb{R}, \mathrm{d}m)$, one obtains

$$\mathcal{V} \circ \mathbf{W}_\lambda \circ \mathcal{V}^* = -s^2 + 2\pi\lambda^2(4N+1) - \frac{1}{4} \tag{1.6}$$

where s^2 is the operator of multiplication by the square of the variable in $L^2(\mathbb{R}, \mathrm{d}m)$. The right hand side of (1.6) is meaningful in the general context of orthogonal polynomials. We compute in Sect. 3.4 the hermitian Jacobi matrix of the scaling operator $\mathbb{S}$ in the basis of orthogonal polynomials. Its diagonal elements are 0, while the non-zero matrix elements above the diagonal are $\mathbb{S}_{n,n+1} = i\sqrt{(n+1/2)(n+1)}$. In Sect. 3.5 we describe the two hermitian Jacobi matrices $J_\pm$ associated to the operators $\mathbb{S}^2$ restricted to the eigenspaces of $\mathbb{F}_{e_\mathbb{R}}$ for the eigenvalues ± 1. Similarly we obtain the Jacobi matrices of the prolate operator $\mathbf{W}_\lambda$.

In Sect. 3.6 we show that the zeros of the zeta function appear from the action of the scaling operator on the quotient by the range of the map $\mathcal{E}$ applied to the positive eigenspace of $\mathbf{W}_\lambda$ when $\lambda \to \infty$.

Section 4 is devoted to the extension of the Hardy–Titchmarsh transform in the semilocal situation involving a finite set S of places of $\mathbb{Q}$ containing the archimedean one, and the analysis of the semilocal Sonin space. The semilocal version of Theorem 3.1 involves a new measure which, up to normalization, is given by the square of the absolute value of the restriction to the critical line of the product of local factors for the places in S.

To state this result we need the following notation (see Sect. 4.1). The ring of semilocal adeles is the product $\mathbb{A}_S = \prod_S \mathbb{Q}_v$, the semilocal adele class space is the quotient $X_S := \mathbb{A}_S/\Gamma$, where

$$\Gamma = \{\pm p_1^{n_1} \cdots p_k^{n_k}\ :\ p_j \in S \setminus \{\infty\},\ n_j \in \mathbb{Z}\} \subset \mathrm{GL}_1(\mathbb{A}_S) = \prod_{p\in S} \mathrm{GL}_1(\mathbb{Q}_p).$$

The semilocal idele class group $C_S := \mathrm{GL}_1(\mathbb{A}_S)/\Gamma$ acts on X_S by multiplication and this action restricts to its maximal compact subgroup K_S. One has a canonical isomorphism w_S of the Hilbert space $L^2(X_S)^{K_S}$ with $L^2(\mathbb{R}_+^*, \mathrm{d}^*u)$, see (4.7). With $\mathbb{F}_\mu : L^2(\mathbb{R}_+^*, \mathrm{d}^*u) \to L^2(\mathbb{R})$ being the Fourier transform for the multiplicative group $\mathbb{R}_+^*$ see (3.4), we obtain the unitary $\mathcal{U}_S := \mathbb{F}_\mu \circ w_S$. Finally we let $\mathcal{M}_S : L^2(\mathbb{R}) \to L^2(\mathbb{R}, \mathrm{d}m_S)$ be the unitary given by

$$\mathcal{M}_S(f)(s) := \left(\prod_{v\in S} L_v(\frac{1}{2} - is)\right)^{-1} f(s),\quad \mathrm{d}m_S(s) := \left|\prod_{v\in S} L_v(\frac{1}{2} - is)\right|^2 \mathrm{d}s. \tag{1.7}$$

where the L_v are the local Euler factors. We then derive the following result

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

Theorem 1 (Proposition 4.1 and Proposition 4.2) *Let S be a finite set of places with $\infty \in S$, let R_S be the maximal compact subring of $\prod_{p\in S\setminus\{\infty\}} \mathbb{Q}_p$, and $f \in L^2(\mathbb{R})^{ev}$. Then let $\eta_S(f)$ be the class of the function $1_{R_S} \otimes f$ in $L^2(X_S)^{K_S}$.*

(i) *One has*

$$\mathbb{F}_\mu w_S(\eta_S(f))(s) = \left(\prod_{p\in S\setminus\{\infty\}} L_p\left(\frac{1}{2} - is\right)\right)(\mathbb{F}_\mu w_\infty f)(s). \tag{1.8}$$

(ii) *Let the Fourier transform for $\mathbb{Q}_p$ be normalized so that the function $1_{\mathbb{Z}_p}$ is its own Fourier transform and let $\mathbb{F}_S$, acting in $L^2(X_S)$, be induced by the tensor product of the local Fourier transforms. One has*

$$\mathbb{F}_S \circ \eta_S = \eta_S \circ \mathbb{F}_{e_\mathbb{R}}.$$

(iii) *The unitary transformation $\mathcal{V}_S := \mathcal{M}_S \circ \mathcal{U}_S : L^2(X_S)^{K_S} \to L^2(\mathbb{R}, dm_S)$ gives the canonical form of the cyclic pair (D, ξ) where $D := \mathbb{S}$ and $\xi = \xi_S := \eta_S(\xi_\infty)$.*

(iv) *The cyclic pair $(\mathbb{S}, \xi_S)$ is even and the $\mathbb{Z}/2\mathbb{Z}$ grading is given by the Fourier transform $\mathbb{F}_S$ which becomes the symmetry $s \mapsto -s$ under the unitary transformation $\mathcal{V}_S$.*

The computation of the coefficients of the hermitian Jacobi matrix of the cyclic pair for a general S as above is deferred to a forthcoming paper. In order to analyse the semilocal Sonin spaces we introduce the dual Hardy–Titchmarsh transform in Sect. 4.4. In the case of the single archimedan place this transform was introduced by J.-F. Burnol in connection with the work of L. de Branges. The main result of this section is the construction of a hilbertian[3] isomorphism $\theta_S : L^2(\mathbb{R})^{ev} \to L^2(X_S)^{K_S}$ which connects together the semilocal Sonin spaces as detailed in the following theorem:

Theorem 2 *Let S be a finite set of places with $\infty \in S$ and $\lambda > 0$. Then the map θ_S induces a hilbertian isomorphism of the Sonin spaces $\theta_S : \mathbf{S}_\lambda(\mathbb{R}, e_\infty) \to \mathbf{S}_\lambda(X_S, \alpha)$ where α is the normalized character.*

This theorem suggests that the resulting invariantly defined Sonin space should, when suitably equipped with a prolate type operator, play the role of the sought-for Weil cohomology. In fact we observed in [10] that, in the case of the single archimedean place, the Sonin space corresponds (up to a finite-dimensional possible discrepancy) to the negative part of the spectrum of the prolate operator and this fact gives another constraint in the search of the semilocal analogue of the prolate operator. Moreover the results of [13] suggest a strong relation between orthogonal polynomials-which play a key role in the semilocal case- and reproducing kernels commuting with differential operators.

[3] We use the term "hilbertian" to denote the underlying topological vector space structure of a Hilbert space.

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

Another conceptual point of view on the prolate operator is obtained in Sect. 5. We show that the prolate operator admits (still at the formal algebraic level) a description as an element of the enveloping algebra of the Lie algebra of $SL_2(\mathbb{R})$ through the metaplectic representation of the double cover of $SL_2(\mathbb{R})$. We compare this description of the prolate operator with Theorem 3.1 in Sect. 5.2. We show that the generators of the metaplectic representation make sense in the context of cyclic pairs i.e. equivalently of orthogonal polynomials. Furthermore, we prove in Theorem 5.2 that the discrepancy coefficients d_n vanish precisely in the situation of Theorem 3.1 and their vanishing determines uniquely the cyclic pair $(\mathbb{S}, \xi_\infty)$, the associated moments and the representation.

In a forthcoming paper we shall develop the role of the Weil representation [21] of the metaplectic cover of the algebraic group $SL_2(\mathbb{A}_S)$ in the semilocal context, thereby introducing a second potential candidate for the semilocal prolate operator.

Appendix 6 is meant to connect our conventions for the Fourier transform on p-adic numbers with those contained in Weil's book [22].

2 Cyclic pairs and associated prolate operators

In this section we give a general construction starting from a pair (D, ξ) of a selfadjoint operator D acting in a Hilbert space $\mathcal{H}$ and a unit vector $\xi \in \cap_{n\in\mathbb{N}} \operatorname{Dom} D^n$. We assume that $\mathcal{H}$ is infinite dimensional and that ξ is a *cyclic vector*, i.e. that the linear span of the vectors $D^j\xi$ for $j \in \mathbb{N}$ is dense in $\mathcal{H}$. Given such a pair we let $E_n \subset \mathcal{H}$ be the linear span of the vectors $D^j\xi$ for $j \leq n$. They are finite dimensional subspaces of $\mathcal{H}$ and $\dim E_n = n + 1$, since the vectors $D^j\xi$ are linearly independent. One has by construction $D(E_n) \subset E_{n+1}$ and the general situation is described as follows by the spectral theorem.

Theorem 2.1 *Let (D, ξ) be a cyclic pair as above.*

(i) *There exists a probability measure μ on $\mathbb{R}$ and a unitary isomorphism $U : \mathcal{H} \to L^2(\mathbb{R}, d\mu)$ which transforms D in the operator of multiplication by the variable $s \in \mathbb{R}$ and the vector ξ in the constant function equal to* 1.
(ii) *Under the isomorphism U the subspace E_n becomes the space of polynomials of degree $\leq n$.*
(iii) *The spectral measure $\int f(s)d\mu(s) := \langle f(D)\xi|\xi\rangle$ is a complete invariant of the cyclic pair.*

Theorem 2.1 gives the canonical form of a cyclic pair (D, ξ). One can then apply the general theory of orthogonal polynomials and obtain an orthonormal basis (ξ_j) where $\xi_0 = \xi$ and E_n is the span of the ξ_j for $j \leq n$. One then considers the *number* operator N which is diagonal in the orthonormal basis (ξ_j) and such that $N\xi_j := j\xi_j$.

Definition 2.2 Let (D, ξ) be a cyclic pair. The *formal prolate* operator $\omega(D, \xi, \lambda)$ is the operator

$$\omega(D, \xi, \lambda) := -D^2 + \lambda^2 N. \tag{2.1}$$

This definition is formal inasmuch as it does not give precisely the domain of the operator. We shall see in Sect. 3 that the standard prolate differential operator $\mathbf{W}_\lambda$ is a

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

special case of Definition 2.2. It is clear in that case that the obtained formal operator is symmetric and finding the relevant selfadjoint extension is delicate.

Definition 2.3 Let (D,ξ) be a cyclic pair. A $\mathbb{Z}/2$-grading γ is a unitary of square 1, $\gamma^2=1$, $\gamma=\gamma^*$ such that $\gamma D=-D\gamma$ and $\gamma\xi=\xi$. A cyclic pair is *even* if it admits a $\mathbb{Z}/2$-grading.

The next proposition gives elementary properties of even cyclic pairs.

Proposition 2.1
1. *A cyclic pair is* even *iff the spectral measure μ of Theorem 2.1 is invariant under $s\mapsto -s$. The $\mathbb{Z}/2$-grading is unique (if it exists) and is equal to $\gamma=\exp(i\pi N)$.*
2. *Let (D,ξ) be an even cyclic pair. The formal prolate operator (2.1) commutes with the $\mathbb{Z}/2$-grading.*

Proof 1. Assume that the measure μ of Theorem 2.1 is invariant under $s\mapsto -s$. Let $\gamma(f)(s):=f(-s)$ for all $f\in L^2(\mathbb{R},d\mu)$. One has $\gamma^2=1$ and γ is unitary by invariance of the measure. By construction γ anticommutes with the multiplication by s and one has $\gamma\xi=\xi$ where $\xi(s)=1$, $\forall s$. Using the unitary isomorphism $U:\mathcal{H}\to L^2(\mathbb{R},d\mu)$ of Theorem 2.1 one obtains a $\mathbb{Z}/2$-grading for (D,ξ). Conversely let γ be a $\mathbb{Z}/2$-grading for (D,ξ). Then the spectral measure is invariant under $s\mapsto -s$, since one has

$$\int h(s)d\mu(s):=\langle h(D)\xi\mid\xi\rangle=\langle h(D)\gamma\xi\mid\gamma\xi\rangle=\langle h(-D)\xi\mid\xi\rangle=\int h(-s)d\mu(s)$$

Let us show the uniqueness of the $\mathbb{Z}/2$-grading. Let γ' be another $\mathbb{Z}/2$-grading. Then the product $u=\gamma\gamma'$ commutes with D and fulfills $u\xi=\xi$. Thus one has $uD^j\xi=D^j\xi$ for all $j\in\mathbb{N}$ and since ξ is cyclic one obtains $u=1$ and hence $\gamma'=\gamma$. Let us show that $\gamma=\exp(i\pi N)$. One has $\gamma D^j\xi=(-1)^jD^j\xi$. This shows that the subspaces E_n (span of the $D^j\xi$ for $j\le n$) are globally invariant under γ and that they decompose as the eigenspaces $E_n^\pm$ generated respectively by the $D^j\xi$ for $j\le n$, j even and j odd. The Gram–Schmidt orthonormalization process thus takes place independently in $E_n^\pm$ yielding the orthonormal basis ξ_j where $\xi_j\in E_n^\pm$ according to the parity of j. It follows that $\gamma=\exp(i\pi N)$.
2. The number operator N commutes with $\gamma=\exp(i\pi N)$, and D^2 also commutes with γ.

2.1 Spectral triple of a cyclic pair

Let (D,ξ) be an even cyclic pair and γ the $\mathbb{Z}/2$-grading. Proposition 2.1 suggests to consider the even spectral triple $(\mathcal{A},\mathcal{H},D)$ where the C^*-algebra $\mathcal{A}:=c_0(\mathbb{N})$ of sequences vanishing at ∞ acts as follows in the Hilbert space $\mathcal{H}$ of the cyclic pair:

$$c_0(\mathbb{N})\times\mathcal{H}\ni(f,\eta)\mapsto f(N)\eta\in\mathcal{H} \tag{2.2}$$

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

Proposition 2.2 *Let (D, ξ) be a cyclic pair, and $\mathcal{A} := c_0(\mathbb{N})$ act in $\mathcal{H}$ by (2.2). Then $(\mathcal{A}, \mathcal{H}, D)$ is a spectral triple and the commutators $[D, f]$ make sense and are bounded for any $f \in c_c(\mathbb{N}) \subset c_0(\mathbb{N})$.*

Proof Let $f \in c_c(\mathbb{N})$, it is by construction a finite linear combination of the delta functions $\delta_j(k) := 0$ for $k \neq j$ and $\delta_j(j) = 1$. The action (2.2) of δ_j in $\mathcal{H}$ is the rank one operator $|\xi_j\rangle\langle\xi_j|$, and its commutator with D makes sense since $\xi_j \in \operatorname{Dom} D$ and it is equal to the bounded operator $|D\xi_j\rangle\langle\xi_j| - |\xi_j\rangle\langle D\xi_j|$. Thus the commutators $[D, f]$ are finite rank operators for any $f \in c_c(\mathbb{N}) \subset c_0(\mathbb{N})$. ☐

One has $D\xi_j \in E_{j+1}$ for all j and $\langle D\xi_n|\xi_k\rangle = 0$ for $k < n-1$ using $D = D^*$. Thus one has

$$D\xi_n = a_{n-1}\xi_{n-1} + a_n\xi_{n+1}, \quad a_n \neq 0, \tag{2.3}$$

where one can fix by induction the phase of the orthonormal vectors ξ_n in such a way that the non-zero scalars $a_n := \langle D\xi_n|\xi_{n+1}\rangle$ are positive. Let $f \in c_c(\mathbb{N})$, one has $f = \sum f(j)\delta_j$ and

$$\begin{aligned}
[D, f] &= \sum f(j)[D, \delta_j] = \sum f(j)a_j \left(|\xi_{j+1}\rangle\langle\xi_j| - |\xi_j\rangle\langle\xi_{j+1}|\right) \\
&\quad + \sum f(j)a_{j-1} \left(|\xi_{j-1}\rangle\langle\xi_j| - |\xi_j\rangle\langle\xi_{j-1}|\right) \\
&= \sum (f(j) - f(j+1))a_j \left(|\xi_{j+1}\rangle\langle\xi_j| - |\xi_j\rangle\langle\xi_{j+1}|\right) \\
&= \sum \alpha_j \left(|\xi_{j+1}\rangle\langle\xi_j| - |\xi_j\rangle\langle\xi_{j+1}|\right)
\end{aligned}$$

where $\alpha_j := (f(j) - f(j+1))a_j$. Let $T(j) := \left(|\xi_{j+1}\rangle\langle\xi_j| - |\xi_j\rangle\langle\xi_{j+1}|\right)$. One has

$$\begin{aligned}
\left\|\sum \alpha_j T(j)\right\| &\leq \left\|\sum \alpha_{2j} T(2j)\right\| + \left\|\sum \alpha_{2j+1} T(2j+1)\right\| \\
&= \max|\alpha_{2j}| + \max|\alpha_{2j+1}|
\end{aligned}$$

since the matrices of the terms appearing in the rhs are block diagonal. Moreover one has

$$\langle\xi_{k+1}|T(j)\xi_k\rangle = \langle\xi_{k+1}|\xi_{j+1}\rangle\langle\xi_j|\xi_k\rangle - \langle\xi_{k+1}|\xi_j\rangle\langle\xi_{j+1}|\xi_k\rangle = \delta(j-k)$$

so that

$$\langle\xi_{k+1}|\left(\sum \alpha_j T(j)\right)\xi_k\rangle = \alpha_k$$

and one obtains the inequalities

$$\max|\alpha_j| \leq \left\|\sum \alpha_j T(j)\right\| \leq \max|\alpha_{2j}| + \max|\alpha_{2j+1}|$$

which gives

$$\max|(f(j) - f(j+1))a_j| \leq \|[D, f]\| \leq \max_{j \text{ even}} |(f(j) - f(j+1))a_j|$$

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

$$+\max_{j \text{ odd}} |(f(j) - f(j+1))a_j|$$

and

$$\max |(f(j) - f(j+1))a_j| \leq \|[D, f]\| \leq 2 \max |(f(j) - f(j+1))a_j| \quad (2.4)$$

We now compute the spectral distance function on $\mathbb{N}$ associated to the spectral triple $(\mathcal{A}, \mathcal{H}, D)$.

Proposition 2.3 *Let* $\phi(n) := \sum_{0 \leq j \leq n} a_j^{-1} \in \mathbb{R}_+$. *The spectral distance function of the spectral triple* $(\mathcal{A}, \mathcal{H}, D)$ *of Proposition 2.2 is equivalent to the distance* $d(n, m) := |\phi(n) - \phi(m)|$, *more precisely*

$$\frac{1}{2} d(n, m) \leq \sup\{|f(n) - f(m)| \mid \|[D, f]\| \leq 1\} \leq d(n, m) \quad (2.5)$$

Proof By construction one has $a_j > 0$ so that the map ϕ is strictly increasing and $d(n, m) := |\phi(n) - \phi(m)|$ defines a distance function on $\mathbb{N}$. The distance $d(n, m)$ is given by

$$d(n, m) = \sup\{|f(n) - f(m)| \mid \max |(f(j) - f(j+1))a_j| \leq 1\}$$

and thus (2.5) follows from (2.4). □

2.2 Jacobi matrix of the prolate operator of a cyclic pair

At the formal level, i.e. ignoring the issue of selfadjointness of D, an even cyclic pair is determined by the sequence (a_n) which gives the Jacobi matrix of the operator D in the basis (ξ_n) by (2.3) while no longer assuming that the a_n are real.

$$D = \begin{pmatrix} 0 & a_0 & 0 & 0 & 0 & \ldots \\ \overline{a}_0 & 0 & a_1 & 0 & 0 & \ldots \\ 0 & \overline{a}_1 & 0 & a_2 & 0 & \ldots \\ 0 & 0 & \overline{a}_2 & 0 & a_3 & \ldots \\ 0 & 0 & 0 & \overline{a}_3 & 0 & \ldots \\ \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \end{pmatrix}$$

One can then determine a Jacobi matrix representation for the formal prolate operator of Definition 2.2. This operator commutes with the grading γ and thus is the direct sum of its restriction to the eigenspaces $\mathcal{H}^{\pm} := \{\xi \in \mathcal{H} \mid \gamma\xi = \pm\xi\}$. These subspaces are generated by the vectors ξ_n where the parity of n is fixed. Indeed the square of the matrix of D is of the form (for $n = 5$)

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

$$\begin{pmatrix} a_0\overline{a}_0 & 0 & a_0a_1 & 0 & 0 & 0 \\ 0 & a_0\overline{a}_0 + a_1\overline{a}_1 & 0 & a_1a_2 & 0 & 0 \\ \overline{a}_0\overline{a}_1 & 0 & a_1\overline{a}_1 + a_2\overline{a}_2 & 0 & a_2a_3 & 0 \\ 0 & \overline{a}_1\overline{a}_2 & 0 & a_2\overline{a}_2 + a_3\overline{a}_3 & 0 & a_3a_4 \\ 0 & 0 & \overline{a}_2\overline{a}_3 & 0 & a_3\overline{a}_3 + a_4\overline{a}_4 & 0 \\ 0 & 0 & 0 & \overline{a}_3\overline{a}_4 & 0 & \dots \end{pmatrix}$$

and its restrictions to $\mathcal{H}^{\pm}$ are Jacobi matrices of the general form

$$\begin{pmatrix} a_0\overline{a}_0 & a_0a_1 & 0 & 0 \\ \overline{a}_0\overline{a}_1 & a_1\overline{a}_1 + a_2\overline{a}_2 & a_2a_3 & 0 \\ 0 & \overline{a}_2\overline{a}_3 & a_3\overline{a}_3 + a_4\overline{a}_4 & \dots \\ 0 & 0 & \overline{a}_4\overline{a}_5 & \dots \end{pmatrix},$$

$$\begin{pmatrix} a_0\overline{a}_0 + a_1\overline{a}_1 & a_1a_2 & 0 & 0 \\ \overline{a}_1\overline{a}_2 & a_2\overline{a}_2 + a_3\overline{a}_3 & a_3a_4 & 0 \\ 0 & \overline{a}_3\overline{a}_4 & a_4\overline{a}_4 + a_5\overline{a}_5 & \dots \\ 0 & 0 & \overline{a}_5\overline{a}_6 & \dots \end{pmatrix}$$

We now give the matrix description of the formal prolate operator of Definition 2.2.

Proposition 2.4 *The formal prolate operator of Definition 2.2 restricted to $\mathcal{H}^{\pm}$ is represented by the Jacobi matrices $J^{\pm}$ whose matrix coefficients are 0 except those given by*

$$J^{+}_{n,n+1} = \overline{J^{+}_{n+1,n}} = -a_{2n}a_{2n+1}, \quad J^{+}_{n,n} = -a_{2n}\overline{a}_{2n} - a_{2n-1}\overline{a}_{2n-1} + 2n\lambda^2 \tag{2.6}$$

$$J^{-}_{n,n+1} = \overline{J^{-}_{n+1,n}} = -a_{2n+1}a_{2n+2}, \quad J^{-}_{n,n} = -a_{2n}\overline{a}_{2n} - a_{2n+1}\overline{a}_{2n+1} + (2n+1)\lambda^2 \tag{2.7}$$

Proof This follows from Definition 2.2 and the computation of the matrix for the restriction of D^2 to $\mathcal{H}^{\pm}$. □

3 Hardy–Titchmarsh transform: archimedean place

This section contains a detailed proof of the following theorem.

Theorem 3.1 *Let h_n be the normalized Hermite functions.*

(i) *One has $\mathcal{U}(h_0) = 2^{-\frac{3}{4}}\pi^{-\frac{1}{2}}L_\infty(\frac{1}{2} - is)$ where $L_\infty(z) := \pi^{-z/2}\Gamma(z/2)$ is the Euler factor at the archimedean place.*

(ii) *The functions $\mathcal{U}(h_{2n})$ are of the form $\mathcal{U}(h_{2n})(s) = (-1)^n\mathcal{P}_n(s)\mathcal{U}(h_0)(s)$ where the $\mathcal{P}_n(s)$ are polynomials of the same parity as n and which are orthogonal (and normalized) for the probability measure $\mathrm{d}m(s) := (2\pi)^{-\frac{3}{2}}|\Gamma(\frac{1}{4} + \frac{is}{2})|^2\mathrm{d}s$ on $\mathbb{R}$.*

(iii) *The map $\mathcal{M} : L^2(\mathbb{R}) \to L^2(\mathbb{R}, \mathrm{d}m)$, $\mathcal{M}(f)(s) := \mathcal{U}(h_0)^{-1}(s)f(s)$ is a unitary isomorphism.*

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

(iv) *The unitary isomorphism* $\mathcal{M} \circ \mathcal{U} : L^2(\mathbb{R})^{ev} \to L^2(\mathbb{R}, \mathrm{d}m)$ *transforms the Hermite functions* h_{2n} *into the orthogonal polynomials* $(-1)^n \mathcal{P}_n$ *and the operator* $\mathbb{S}$ *into the multiplication by* s.

(v) *Let* N *be the grading operator* $N(\mathcal{P}_n) := n\, \mathcal{P}_n$ *in* $L^2(\mathbb{R}, \mathrm{d}m)$. *One has*

$$(\mathcal{M} \circ \mathcal{U}) \circ \mathbf{W}_\lambda \circ (\mathcal{M} \circ \mathcal{U})^* = -s^2 + 2\pi\lambda^2(4N+1) - \frac{1}{4} \tag{3.1}$$

where s^2 *is the operator of multiplication by the square of the variable in* $L^2(\mathbb{R}, \mathrm{d}m)$.

By construction the Hermite functions h_n are themselves of the form $h_n(x) = H_n(x)h_0(x)$ where the H_n are orthogonal (and normalized) polynomials. Since the Fourier transform $\mathbb{F}_\mu$ does not preserve the product of functions nor their polynomial nature, Theorem 3.1 is by no means a consequence of the construction of the Hermite functions. The operator $\mathcal{U} \circ S \circ \mathcal{U}^*$ is the multiplication by s while the Hermite operator gives the grading. The main result is that, due to the functional equation $\Gamma(x+1) = x\Gamma(x)$ of the Γ-function one has natural orthogonal polynomials governing the functions $\mathcal{U}(h_{2n})$ where the h_n are the normalized Hermite functions.

3.1 Notation

3.1.1 $\mathbb{F}_{e_\mathbb{R}} : L^2(\mathbb{R})^{ev} \to L^2(\mathbb{R})^{ev}$

The Fourier transform $\mathbb{F}_{e_\mathbb{R}} : L^2(\mathbb{R}) \to L^2(\mathbb{R})$

$$\mathbb{F}_{e_\mathbb{R}}(f)(y) := \int f(x)\exp(-2\pi i x y)\mathrm{d}x \tag{3.2}$$

restricts to the subspace $L^2(\mathbb{R})^{ev} \subset L^2(\mathbb{R})$ of even functions.

3.1.2 $w_\infty : L^2(\mathbb{R})^{ev} \to L^2(\mathbb{R}_+^*, \mathrm{d}^*u)$

The Haar measure of the multiplicative group $\mathbb{R}_+^*$ of positive real numbers is $\mathrm{d}^*u := \mathrm{d}u/u$. The map w_∞ is defined by

$$w_\infty(f)(u) := u^{1/2} f(u)\,, \ \ \forall u \in \mathbb{R}_+^*. \tag{3.3}$$

3.1.3 $\mathbb{F}_\mu : L^2(\mathbb{R}_+^*, \mathrm{d}^*u) \to L^2(\mathbb{R})$

The Fourier transform for the multiplicative group $\mathbb{R}_+^*$ is

$$\mathbb{F}_\mu(f)(s) := \int_0^\infty f(u)\, u^{-is}\mathrm{d}^*u. \tag{3.4}$$

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

3.1.4 $\mathcal{U} : L^2(\mathbb{R})^{ev} \to L^2(\mathbb{R})$

The unitary transformation $\mathcal{U} := \pi^{-\frac{1}{2}}\, \mathbb{F}_\mu \circ w_\infty$ is given by

$$\mathcal{U} : L^2(\mathbb{R})^{ev} \to L^2(\mathbb{R}),\;\; \mathcal{U}(f)(s) = \pi^{-\frac{1}{2}} \int_0^\infty f(v) v^{\frac{1}{2}-is} d^*v\,. \tag{3.5}$$

3.2 Canonical form of the scaling operator

We let $H := x\partial_x$ and $\mathbb{S} := -i(H + \frac{1}{2})$ acting in the Hilbert space $L^2(\mathbb{R})^{ev}$ as unbounded operators. By construction $\mathbb{S}$ is the selfadjoint generator of the scaling unitary transformations

$$(\exp(it\mathbb{S})f)(x) = \lambda^{1/2} f(\lambda x),\;\; \lambda = e^{t/2},\;\; \forall t \in \mathbb{R}.$$

We let $\xi_\infty = h_0 \in L^2(\mathbb{R})^{ev}$ be the vector of norm 1 given by

$$h_0(x) := 2^{\frac{1}{4}} \exp(-\pi x^2)\,,\;\; \forall x \in \mathbb{R}.$$

One has

$$2^{-\frac{1}{4}} \pi^{\frac{1}{2}}\, \mathcal{U}(h_0)(s) = \int_0^\infty v^{1/2-is} e^{-\pi v^2} d^*v = \frac{1}{2}\pi^{-\frac{1}{4}+i\frac{s}{2}} \Gamma\left(\frac{1}{4} - i\frac{s}{2}\right)$$

We define the unitary transformation

$$\mathcal{M} : L^2(\mathbb{R}) \to L^2(\mathbb{R}, dm),\;\; \mathcal{M}(f)(s) := \mathcal{U}(h_0)^{-1}(s) f(s) \tag{3.6}$$

where the measure dm is $|\mathcal{U}(h_0)|^2 ds$.

Proposition 3.1 (i) *The unitary transformation $\mathcal{V} := \mathcal{M} \circ \mathcal{U} : L^2(\mathbb{R})^{ev} \to L^2(\mathbb{R}, dm)$ gives the canonical form of the cyclic pair $(\mathbb{S}, \xi_\infty))$ where $\mathbb{S} := -i(H + \frac{1}{2})$ and $\xi_\infty = h_0$.*

(ii) *The cyclic pair $(\mathbb{S}, \xi_\infty)$ is even and the (unique) grading γ is equal to the Fourier transform $\mathbb{F}_{e_\mathbb{R}} : L^2(\mathbb{R})^{ev} \to L^2(\mathbb{R})^{ev}$.*

Proof (i) The conjugate of $\mathbb{S}$ by $\mathcal{U}$ is the multiplication by the variable s since

$$\int_0^\infty ((H + \frac{1}{2})f)(v) v^{\frac{1}{2}-is} d^*v = \int_0^\infty \partial_v (v^{\frac{1}{2}} f(v)) v^{-is} dv = is \int_0^\infty f(v) v^{\frac{1}{2}-is} d^*v$$

This remains true when conjugating by $\mathcal{V}$, while by construction $\mathcal{V}(\xi_\infty)$ is the constant function $\mathcal{V}(\xi_\infty)(s) = 1$.

(ii) The vector $\xi_\infty = h_0$ is invariant under $\mathbb{F}_{e_\mathbb{R}}$ and this transformation is unitary of square 1 and anticommutes with $\mathbb{S}$, thus the result follows from Proposition 2.1. ◻

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

3.3 Number operator *N*

We relate the number operator N of the cyclic pair $(\mathbb{S}, \xi_\infty)$ with the Hermite differential operator $\mathbf{H} = -\partial^2 + (2\pi q)^2$.

Proposition 3.2 (i) *The number operator N of the cyclic pair $(\mathbb{S}, \xi_\infty)$ is given by*

$$N = \frac{\mathbf{H}}{8\pi} - \frac{1}{4}, \quad \mathbf{H} = -\partial^2 + (2\pi q)^2 \tag{3.7}$$

(ii) *The eigenfunctions of N are the even Hermite functions $h_{2n} \in L^2(\mathbb{R})^{ev}$ where*

$$h_{2n}(x) = \sum_0^n (-1)^{n-k} 2^{-n+3k+\frac{1}{4}} \frac{((2n)!)^{\frac{1}{2}}}{(2k)!(n-k)!} \pi^k x^{2k} e^{-\pi x^2} \tag{3.8}$$

Proof The Hermite functions (3.8) are eigenfunctions for the operator $\mathbf{H}$ and give the orthonormal basis of eigenfunctions for the restriction of $\mathbf{H}$ to $L^2(\mathbb{R})^{ev}$ and the eigenvalues are $2\pi(4n+1)$. The subspace $E_n \subset L^2(\mathbb{R})^{ev}$ linear span of the vectors $\mathbb{S}^j \xi_\infty$ for $j \leq n$ is formed of the even functions $e^{-\pi x^2} P(x)$ where $P(x)$ is an even polynomial of degree $\leq 2n$. It coincides with the linear span of the functions h_{2j}, $0 \leq j \leq n$. Thus one obtains (i) and (ii). $\square$

As a corollary we get Eq. (3.1) of Theorem 3.1.

Corollary 3.2

$$(\mathcal{M} \circ \mathcal{U}) \circ \mathbf{W}_\lambda \circ (\mathcal{M} \circ \mathcal{U})^* = -s^2 + 2\pi\lambda^2(4N+1) - \frac{1}{4} \tag{3.9}$$

Proof One has by (1.5)

$$\begin{aligned}(\mathbf{W}_\lambda \psi)(q) &= \partial(q^2 \partial\, \psi(q)) + \lambda^2\, \mathbf{H}\, \psi(q) \\ &= \left(-\mathbb{S}^2 + \lambda^2\, \mathbf{H} - \frac{1}{4}\right) \psi(q)\end{aligned}$$

which gives (3.9) using (3.7) since $\mathcal{V} \circ \mathbb{S} \circ \mathcal{V}^*$ is given by multiplication by s by Proposition 3.1. $\square$

3.4 Jacobi matrix for $\mathbb{S}$

This section should be compared, for the formulas of the tridiagonal matrix, with [15, Section 8] (formulas based on Hermite series).

We let

$$\mathcal{P}_m(x) := \sqrt{(2m)!} \sum_0^m (-1)^k 2^{3k-m} \frac{\prod_{j=0}^{k-1} \left(j - \frac{ix}{2} + \frac{1}{4}\right)}{(2k)!(m-k)!} \tag{3.10}$$

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

Proposition 3.3 (i) *The polynomials* $\mathcal{P}_m$ *are orthonormal polynomials with respect to the probability measure* $(2\pi)^{-3/2}|\Gamma\left(\frac{1}{4}-i\frac{x}{2}\right)|^2 dx$ *on the line.*

(ii) *The matrix acting on column vectors*[4] *of the operator of multiplication by* x *in the basis of the* $\mathcal{P}_m$ *is the hermitian Jacobi matrix with* 0 *on the diagonal and* $\alpha_n = \frac{i}{2}\sqrt{(2n+1)(2n+2)}$ *in the upper diagonal.*

Proof (i) The transform $\mathcal{V}(h_{2m})$ by the unitary $\mathcal{V} := \mathcal{M} \circ \mathcal{U} : L^2(\mathbb{R})^{ev} \to L^2(\mathbb{R}, dm)$ is given by the polynomial $(-1)^m \mathcal{P}_m$ as one gets using the equality

$$\int_0^\infty x^{1/2-is}\pi^k x^{2k} e^{-\pi x^2} d^*x = \frac{1}{2}\pi^{-\frac{1}{4}+i\frac{s}{2}}\Gamma\left(\frac{1}{4}-i\frac{s}{2}+k\right) \tag{3.11}$$

and the functional equation for $\Gamma(z)$. The polynomial $\mathcal{P}_0 = 1$ gives the normalization. The orthonormality then follows from unitarity of $\mathcal{V}$.

(ii) Using $\mathbb{F}_\mu \circ w$ the operator $H + \frac{1}{2}$ corresponds to the multiplication by ix, thus it is enough to determine the coefficient of $x^{2n+2}e^{-\pi x^2}$ in $(H+\frac{1}{2})h_{2n}$. The term of highest degree in h_{2n} is

$$2^{2n+\frac{1}{4}}(2n)!^{-\frac{1}{2}}\pi^n x^{2n} e^{-\pi x^2}$$

When applying the operator $H = x\partial_x$ to $e^{-\pi x^2}x^k$ one gets

$$H(e^{-\pi x^2}x^k) = e^{-\pi x^2}x^k\left(-2\pi x^2 + k\right)$$

Thus the coefficient of $x^{2n+2}e^{-\pi x^2}$ in $(H+\frac{1}{2})h_{2n}$ is $(-2\pi)\, 2^{2n+\frac{1}{4}}(2n)!^{-\frac{1}{2}}\pi^n$ while in h_{2n+2} it is $2^{2n+2+\frac{1}{4}}(2n+2)!^{-\frac{1}{2}}\pi^{n+1}$. Thus for this highest term one has $(H+\frac{1}{2})h_{2n} \sim c_n h_{2n+2}$ for

$$\begin{aligned} c_n &= (-2\pi)\, 2^{2n+\frac{1}{4}}(2n)!^{-\frac{1}{2}}\pi^n / \left(2^{2n+2+\frac{1}{4}}(2n+2)!^{-\frac{1}{2}}\pi^{n+1}\right) \\ &= -\frac{1}{2}\sqrt{(2n+1)(2n+2)} \end{aligned}$$

which gives, using the orthonormal property of the h_n, the equality

$$\left\langle \left(H+\frac{1}{2}\right)h_{2n}, h_{2n+2}\right\rangle = -\frac{1}{2}\sqrt{(2n+1)(2n+2)} \tag{3.12}$$

To check the signs one takes the simplest case, one has

$$\left(\left(H+\frac{1}{2}\right)h_0\right)(x) = (2^{\frac{1}{4}-1} - 2^{\frac{1}{4}+1}\pi x^2)e^{-\pi x^2}, \quad h_2(x) = \frac{4\pi x^2 - 1}{\sqrt[4]{2}}$$

[4] The matrix with entries $\langle x\mathcal{P}_n, \mathcal{P}_m\rangle$ is the transposed matrix.

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

so that $(H+\frac{1}{2})h_0=-\frac{1}{\sqrt{2}}h_2$. For $m=1$ one has $\mathcal{P}_1(x)=i\sqrt{2}\,x$. Thus $x\mathcal{P}_0=-\frac{i}{\sqrt{2}}\mathcal{P}_1$. From (3.12) we thus obtain that the matrix of the multiplication by x, acting on column vectors is of the form

$$\begin{pmatrix} 0 & \frac{i}{\sqrt{2}} & 0 & 0 & 0 & 0 \\ -\frac{i}{\sqrt{2}} & 0 & i\sqrt{3} & 0 & 0 & 0 \\ 0 & -i\sqrt{3} & 0 & i\sqrt{\frac{15}{2}} & 0 & 0 \\ 0 & 0 & -i\sqrt{\frac{15}{2}} & 0 & i\sqrt{14} & 0 \\ 0 & 0 & 0 & -i\sqrt{14} & 0 & 3i\sqrt{\frac{5}{2}} \\ 0 & 0 & 0 & 0 & -3i\sqrt{\frac{5}{2}} & 0 \end{pmatrix}$$

i.e. that one has for all n

$$x\mathcal{P}_n=\overline{\alpha_n}\,\mathcal{P}_{n+1}+\alpha_{n-1}\mathcal{P}_{n-1},\quad \alpha_n=\frac{i}{2}\sqrt{(2n+1)(2n+2)} \tag{3.13}$$

so that the component of $x\mathcal{P}_n$ on $\mathcal{P}_{n+1}$ is $-\frac{i}{2}\sqrt{(2n+1)(2n+2)}$. □

We show in Proposition 3.4 below that for the cyclic pair corresponding to the archimedean place, the spectral distance is equivalent to the metric induced on $\mathbb{N}^*\subset\mathbb{R}_+^*$ by the invariant metric of the Lie group $\mathbb{R}_+^*$.

Proposition 3.4 *The spectral metric for the cyclic pair $(\mathbb{S},\xi_\infty)$ of Proposition 3.1 is equivalent to the metric induced on $\mathbb{N}^*\subset\mathbb{R}_+^*$ by the invariant metric of the Lie group $\mathbb{R}_+^*$.*

Proof The sequence (a_n) of (2.3), obtained after a change of phase in the orthonormal basis, is given by $a_n=\sqrt{(n+\frac{1}{2})(n+1)}$. For $n>0$ one has

$$1\le a_n\,\log\left(1+\frac{1}{n}\right)\le\sqrt{3}\log 2$$

which shows that the metric induced on $\mathbb{N}^*\subset\mathbb{R}_+^*$ by the invariant metric of the Lie group $\mathbb{R}_+^*$ is equivalent to the metric $d(n,m):=|\phi(n)-\phi(m)|$ of Proposition 2.3. In turns this latter metric is equivalent to the spectral metric by Proposition 2.3. □

3.5 Jacobi matrix of the prolate operator

We now determine the two Jacobi matrices given by the restrictions $\mathbf{W}_\lambda^\pm$ of the prolate operator (1.3) to the even and odd subspaces $\mathcal{H}^\pm$ as in Proposition 2.4. In fact one could compute directly these Jacobi matrices in the basis of Hermite functions and compare with [15, Subsection 8.5.2] (formulas based on Hermite series).

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

The Jacobi matrix $\mathbf{W}_\lambda^+$ is of the following form

$$\begin{pmatrix} 2\pi\lambda^2-\frac{3}{4} & \sqrt{\frac{3}{2}} & 0 & 0 & 0 \\ \sqrt{\frac{3}{2}} & 18\pi\lambda^2-\frac{43}{4} & \sqrt{105} & 0 & 0 \\ 0 & \sqrt{105} & 34\pi\lambda^2-\frac{147}{4} & 3\sqrt{\frac{165}{2}} & 0 \\ 0 & 0 & 3\sqrt{\frac{165}{2}} & 50\pi\lambda^2-\frac{315}{4} & \sqrt{2730} \\ 0 & 0 & 0 & \sqrt{2730} & 66\pi\lambda^2-\frac{547}{4} \end{pmatrix}$$

For a Jacobi matrix J, we use the notation a_n, $n \geq 0$ for the coefficients $J_{n,n+1} = \overline{J_{n+1,n}}$, and b_n, $n \geq 0$ for the diagonal, i.e. $b_n = J_{n,n}$.

Proposition 3.5 *The coefficients of the Jacobi matrix of the prolate operator $\mathbf{W}_\lambda^+$ are given by*

$$a_n = \frac{1}{4}\left((4n+1)(4n+2)(4n+3)(4n+4)\right)^{1/2}, \; b_n = -8n^2-2n-\frac{3}{4}+2\pi\lambda^2(8n+1) \tag{3.14}$$

In the odd case, i.e. for $\mathbf{W}_\lambda^-$, they are

$$a_n = \frac{1}{4}\left((4n+3)(4n+4)(4n+5)(4n+6)\right)^{1/2},$$

$$b_n = -8n^2-10n-\frac{15}{4}+2\pi\lambda^2(8n+5) \tag{3.15}$$

Proof This follows from (3.9) combined with Proposition 2.4 applied to the coefficients $\alpha_n = \frac{i}{2}\sqrt{(2n+1)(2n+2)}$ determined in Proposition 3.3. □

3.6 Link with the map $\mathcal{E}$ and zeta

In [9] we exhibited minuscule eigenvalues of the quadratic form associated to the Weil explicit formulas restricted to test functions supported in a fixed compact interval. The corresponding eigenfunctions were constructed using the image through the map $\mathcal{E}$ of the positive spectrum of the prolate operator $\mathbf{W}_\lambda$. We used these functions to condition the canonical spectral triple of the circle in such a way that they belong to the kernel of the perturbed Dirac operator and obtained the low lying zeros of the Riemann zeta function. For $\lambda \to \infty$ the eigenfunctions for positive eigenvalues of $\mathbf{W}_\lambda$ are approximated by the eigenfunctions of the operator $\mathbf{H} = -\partial^2 + (2\pi q)^2$. The latter eigenfunctions are the Hermite functions $\{h_{2n}\}$.

Following [6, Proposition 2.24] we now explain in which sense the conditioning of the scaling operator by $\mathcal{E}(\{h_{2n}\})$, performed by taking a quotient (in place of an orthogonal complement as above) relates to the zeros of zeta.

Let $\mathcal{S}(\mathbb{R})_0^{\mathrm{ev}}$ be the subspace of the even part of the Schwartz space $\mathcal{S}(\mathbb{R})$ obtained by imposing the two conditions $f(0) = \widehat{f}(0) = 0$. We describe simple linear combinations of the functions h_{2m} which belong to $\mathcal{S}(\mathbb{R})_0^{\mathrm{ev}}$ and then compute their image under the map $\mathbb{F}_\mu \circ \mathcal{E}$, where

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

$$\mathcal{E} := w_\infty \circ \Sigma, \quad \Sigma(f)(x) := \sum_{\mathbb{N}} f(nx).$$

The Fourier transform $\mathbb{F}_{e_\mathbb{R}}(h_{2m})$ is $(-1)^m h_{2m}$ and thus the two conditions $f(0) = \widehat{f}(0) = 0$ are fulfilled by the two families of functions

$$\psi_\ell^+ := h_{4\ell} - \frac{h_{4\ell}(0)}{h_0(0)} h_0, \quad \psi_\ell^- := -h_{4\ell+2} + \frac{h_{4\ell+2}(0)}{h_2(0)} h_2 \tag{3.16}$$

One has $h_{2n}(0) = (-1)^n \frac{2^{\frac{1}{4}-n}\sqrt{(2n)!}}{n!}$. Thus one gets

$$\psi_\ell^+(x) = \sum_1^{2\ell} (-1)^k 2^{-2\ell+3k+\frac{1}{4}} \frac{((4\ell)!)^{\frac{1}{2}}}{(2\ell-k)!(2k)!} \pi^k x^{2k} e^{-\pi x^2} \tag{3.17}$$

In the odd case one has $h_2(x)/h_2(0) = e^{-\pi x^2}\left(4\pi x^2 - 1\right)$. The coefficient of $\pi x^2 e^{-\pi x^2}((4\ell+2)!)^{\frac{1}{2}}$ in $h_{4\ell+2} - \frac{h_{4\ell+2}(0)}{h_2(0)} h_2$ is

$$2^{3-2\ell-1+\frac{1}{4}} \frac{1}{(2\ell)!2!} - 4 \times 2^{-2\ell-1+\frac{1}{4}} \frac{1}{(2\ell+1)!} = 2^{-2\ell+\frac{1}{4}} \frac{4\ell}{(2\ell+1)!}$$

thus one has

$$\psi_\ell^-(x) = \left(-\frac{\ell\, 2^{-2\ell+\frac{9}{4}}((4\ell+2)!)^{\frac{1}{2}}}{(2\ell+1)!} \pi x^2 + \sum_2^{2\ell+1} (-1)^k 2^{3k-2\ell-1+\frac{1}{4}} \frac{((4\ell+2)!)^{\frac{1}{2}}}{(2\ell+1-k)!(2k)!} \pi^k x^{2k} \right) e^{-\pi x^2} \tag{3.18}$$

We now relate the Fourier transform $\mathbb{F}_\mu(w_\infty(\psi_\ell^\pm))$ to the polynomials (3.10).

Lemma 3.3 (i) *The Fourier transform $\mathbb{F}_\mu(w_\infty(\psi_\ell^\pm))$ is equal to the product of the archimedean local factor $\pi^{-\frac{z}{2}}\Gamma\left(\frac{z}{2}\right)$ at $z = \frac{1}{2} - is$, by the following polynomials*

$$P_\ell^+(s) := \sum_1^{2\ell} (-1)^k 2^{-2\ell+3k-\frac{3}{4}} \frac{((4\ell)!)^{\frac{1}{2}}}{(2\ell-k)!(2k)!} \prod_0^{k-1} \left(j - \frac{i\,s}{2} + \frac{1}{4} \right) \tag{3.19}$$

which is even, divisible by $\frac{1}{4} + s^2$, and with real coefficients and

$$\begin{aligned} P_\ell^-(s) := &-\frac{\ell\, 2^{-2\ell+\frac{5}{4}}((4\ell+2)!)^{\frac{1}{2}}}{(2\ell+1)!} \left(-\frac{is}{2} + \frac{1}{4} \right) + \\ &+ \sum_2^{2\ell+1} (-1)^k 2^{3k-2\ell-\frac{7}{4}} \frac{((4\ell+2)!)^{\frac{1}{2}}}{(2\ell+1-k)!(2k)!} \prod_0^{k-1} \left(j - \frac{is}{2} + \frac{1}{4} \right) \end{aligned} \tag{3.20}$$

which is odd, divisible by $\frac{1}{4} + s^2$, and with purely imaginary coefficients.

Journal: **43034** Article No.: **0388** TYPESET DISK LE CP Disp.:**2024/9/12** Pages: **38** Layout: **Small-Ex**

(ii) *One has*

$$P_\ell^+(s) = 2^{-\frac{3}{4}}\left(\mathcal{P}_{2\ell}(s) - \mathcal{P}_{2\ell}(\frac{i}{2})\right), \quad P_\ell^-(s) = 2^{-\frac{3}{4}}\left(\mathcal{P}_{2\ell+1}(s) + 2i\,s\,\mathcal{P}_{2\ell+1}\left(\frac{i}{2}\right)\right) \tag{3.21}$$

Proof (i) The formulas (3.19) and (3.20) follow directly from (3.17) and (3.18) together with (3.11). The divisibility by $\frac{1}{4}+s^2$ follows from the equality

$$\mathbb{F}_\mu(w(\psi_\ell^\pm))\left(\frac{i}{2}\right) = \int_0^\infty x\,\psi_\ell^\pm(x)\mathrm{d}^*x = \frac{1}{2}\int_{-\infty}^\infty \psi_\ell^\pm(x)\mathrm{d}x = 0$$

It follows from (3.17) and (3.18) that $P_\ell^\pm(i/2) = 0$ and the parity of the polynomial $P_\ell^\pm$ then shows that it is divisible by $\frac{1}{4}+s^2$.
(ii) This follows since (3.21) uniquely specifies the required correction of the rescaled polynomials $\mathcal{P}_m$ to make them divisible by $\frac{1}{4}+s^2$ without altering their parity. □

The Riemann–Landau Ξ function

$$\Xi(s) = \frac{z(z-1)}{2}\Gamma(z/2)\pi^{-z/2}\zeta(z), \quad z = \frac{1}{2}+is \tag{3.22}$$

is entire, of Hadamard order one, even and real-valued for real s, and one has

$$\Xi(s) = \Xi(0)\prod\left(1-\frac{s^2}{\alpha^2}\right), \quad \Xi(0) = -\frac{\zeta\left(\frac{1}{2}\right)\Gamma\left(\frac{1}{4}\right)}{8\sqrt[4]{\pi}} \sim 0.497121$$

where $\frac{1}{2}+i\alpha$ runs through the zeros of zeta with positive imaginary part.

Let then $P_\ell^\pm(s) = -\frac{1}{2}(\frac{1}{4}+s^2)R_\ell^\pm(s)$ define the polynomials $R_\ell^\pm(s)$. The next proposition shows that these polynomials give the factorization of $\mathbb{F}_\mu(\mathcal{E}(\psi_\ell^\pm))$ as a multiple of the Ξ function, and that all polynomial multiples of Ξ are obtained in this way. We let $\mathcal{H}_{\le 1}$ be the Hadamard topological ring of entire functions of order ≤ 1.

Proposition 3.6 (i) *One has the equality*

$$\mathbb{F}_\mu(\mathcal{E}(\psi_\ell^\pm))(s) = R_\ell^\pm(s)\,\Xi(s)$$

(ii) *The spectrum of the operator of multiplication by s in the quotient of $\mathcal{H}_{\le 1}$ by the closure of the subspace spanned by the $\mathbb{F}_\mu(\mathcal{E}(\psi_\ell^\pm))$ is the set of $s \in \mathbb{C}$ such that $\frac{1}{2}+is$ is a non-trivial zero of zeta.*

Proof This follows from Lemma 3.3, as in [6, Proposition 2.24]. □

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

4 Semilocal case

4.1 Notations

Let S be a finite set of places, $\infty \in S$, and $\mathbb{A}_S$ the locally compact ring

$$\mathbb{A}_S = \prod_{v \in S} \mathbb{Q}_v. \tag{4.1}$$

This ring contains $\mathbb{Q}$ as a subring using the diagonal embedding. Let $\mathbb{Q}_S$ denote the subring of $\mathbb{Q}$ given by rational numbers whose denominator only involves primes $p \in S$. In other words,

$$\mathbb{Q}_S = \{q \in \mathbb{Q} \mid |q|_v \leq 1\,,\ \forall v \notin S\}. \tag{4.2}$$

The group $\Gamma := \mathbb{Q}_S^*$ of invertible elements of the ring $\mathbb{Q}_S$ is of the form

$$\Gamma = \mathrm{GL}_1(\mathbb{Q}_S) = \{\pm p_1^{n_1} \cdots p_k^{n_k}\ :\ p_j \in S \setminus \{\infty\}\,,\ n_j \in \mathbb{Z}\}. \tag{4.3}$$

The semilocal adèle class space X_S is by definition the quotient

$$X_S := \mathbb{A}_S / \Gamma, \tag{4.4}$$

and we let $\pi_S : \mathbb{A}_S \to X_S$ be the canonical projection. The module extends to a multiplicative map $| \bullet |_S$ from the ring $\mathbb{A}_S = \prod_{v \in S} \mathbb{Q}_v$ to $\mathbb{R}_+$

$$|(u_v)_{v \in S}|_S = \prod |u|_v \in \mathbb{R}_+ \tag{4.5}$$

and by construction this map passes to the quotient $X_S = \mathbb{A}_S / \Gamma$. The groups

$$\mathrm{GL}_1(\mathbb{A}_S) = \prod_{p \in S} \mathrm{GL}_1(\mathbb{Q}_p), \quad C_S = \mathrm{GL}_1(\mathbb{A}_S)/\Gamma \tag{4.6}$$

act naturally by multiplication on the quotient X_S and the orbit of $1 \in \mathbb{A}_S$ gives an embedding $C_S \to X_S$. The complement of C_S in X_S is of measure zero for the product of the Haar measures of the additive groups of the local fields (which is preserved by the action of the countable group Γ). Using the Radon–Nikodym derivative of the Haar measures of the multiplicative groups with respect to the Haar measure of the additive groups, one obtains a unitary identification

$$w_S : L^2(X_S) \to L^2(C_S) \tag{4.7}$$

(see [6, Proposition 2.30]). We also recall (see [6, Eqs. (2.223) and (2.239)] that C_S is a modulated locally compact group with module

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

$$\mathrm{Mod}_S(\lambda) = |\lambda|_S := \prod_{p\in S} |\lambda_p|, \quad \forall \lambda = (\lambda_p) \in C_S \tag{4.8}$$

which is (non-canonically) isomorphic to $\mathbb{R}_+^* \times K_S$, where K_S is the kernel of Mod_S.

4.2 Semilocal Hardy–Titchmarsh transform

We extend the formulas (3.5) and (3.6) to the semilocal case. We use the notation of Sect. 4.1.

Let S be a finite set of places, $\infty \in S$, let R_S be the maximal compact subring of $\prod_{p\neq\infty} \mathbb{Q}_p$. For $f \in L^2(\mathbb{R})^{ev}$ let $\eta_S(f)$ be the class of the function $1_{R_S} \otimes f$ in $L^2(X_S)$. More precisely one has, ignoring the summation over ± 1 since f is even, and letting $\Gamma_+ := \Gamma \cap \mathbb{Q}_+$,

$$\eta_S(f)(x) := \sum_{\Gamma_+} f(\gamma\, \tilde{x}), \;\; \forall \tilde{x} \mid \pi_S(\tilde{x}) = x. \tag{4.9}$$

The restriction of $\eta_S(f)$ to the fundamental domain $F = R_S^* \times \mathbb{R}_+^*$ of the action of Γ on $\mathrm{GL}_1(\mathbb{A}_S) = \prod_{p\in S} \mathrm{GL}_1(\mathbb{Q}_p)$ is R_S^*-invariant and, since $1_{R_S} \otimes f$ vanishes on $\gamma(1 \times u)$ unless $\gamma \in \mathbb{Z}$. One has

$$\eta_S(f)(1 \times u) = \sum_{\Gamma_+\cap\mathbb{Z}} f(\gamma\, u). \tag{4.10}$$

It follows that

$$w_S(\eta_S(f))(u) = \mathcal{E}_S(f)(u) \tag{4.11}$$

where

$$\mathcal{E}_S(f)(u) := u^{1/2} \sum_{\Gamma_+\cap\mathbb{Z}} f(\gamma\, u) \tag{4.12}$$

Proposition 4.1 *Let $f \in L^2(\mathbb{R})^{ev}$. Then with $L_p(z) = (1 - p^{-z})^{-1}$ for $z \in \mathbb{C}$,*

(i) *One has*

$$\mathbb{F}_\mu(\mathcal{E}_S(f))(s) = \left(\prod_{p\in S\setminus\{\infty\}} L_p\left(\frac{1}{2} - is\right) \right) (\mathbb{F}_\mu w_\infty f)(s). \tag{4.13}$$

(ii) *Let $\lambda > 0$. One has $\eta_S(P_\lambda) = P_\lambda^S$ where P_λ^S is the subspace of $L^2(X_S)^{K_S}$ of functions with support in $\{x \mid |x| \le \lambda\}$.*

(iii) *Let the Fourier transform for $\mathbb{Q}_p$ be normalized so that the function $1_{\mathbb{Z}_p}$ is its own Fourier transform and let $\mathbb{F}_S$, acting in $L^2(X_S)$, be induced by the tensor product of the local Fourier transforms. One has*

$$\mathbb{F}_S \circ \eta_S = \eta_S \circ \mathbb{F}_{e_\mathbb{R}}.$$

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

(iv) *One has* $\eta_S(\widehat{P_\lambda}) = \widehat{P_\lambda^S}$ *where* $\widehat{P_\lambda}$ *and* $\widehat{P_\lambda^S}$ *are the images of* P_λ, P_λ^S *by the Fourier transform.*

Proof (i) For simplicity we consider the case of $S = \{p, \infty\}$. Then one has

$$\mathbb{F}_\mu \mathcal{E}_p(f)(s) = L_p\left(\frac{1}{2} - is\right)(\mathbb{F}_\mu w_\infty f)(s).$$

Indeed, one has

$$\begin{aligned}\mathbb{F}_\mu \mathcal{E}_p(f)(s) &= \int_0^\infty u^{1/2} \sum f(p^k u) u^{-is} d^*u = \sum_{k=0}^\infty \int_0^\infty f(p^k u) u^{\frac{1}{2}-is} d^*u \\ &= \sum_k \int_0^\infty f(v) p^{-\frac{k}{2}} p^{iks} v^{\frac{1}{2}-is} d^*v \\ &= \sum_k p^{-\frac{k}{2}} p^{iks} \int_0^\infty f(v) v^{\frac{1}{2}-is} d^*v \\ &= \int_0^\infty f(v) v^{\frac{1}{2}-is} d^*v \,(1 - p^{-\frac{1}{2}-is})^{-1} \\ &= L_p\left(\frac{1}{2} - is\right)(\mathbb{F}_\mu w_\infty f)(s).\end{aligned}$$

(ii) The module $|x|$ is equal to 1 on principal ideles and hence on elements $\gamma \in \Gamma = \{\pm \prod p_v^{n_v} \mid n_v \in \mathbb{Z}\}$. Thus $|x|$ makes sense on $X_S = \left(\prod \mathbb{Q}_v\right) / \Gamma$ and so does P_λ^S. Let $f \in P_\lambda$ and $\eta_S(f)$ be the class of the function $1_{R_S} \otimes f$ in $L^2(X_S)$. Let $x = (y, u) \in R_S \times \mathbb{R}$ with $|x| > \lambda$. One has

$$\lambda < |x| = |y||u| \leq |u| \Rightarrow |u| > \lambda \Rightarrow f(u) = 0.$$

Conversely, let $g \in P_\lambda^S$. Let $h \in L^2(\mathbb{R})^{ev}$ be such that $w_S(g) = w_\infty(h)$, i.e. that $u^{1/2} g(y \times u) = u^{1/2} h(u)$ for any $y \in R_S^*$, $u > 0$. One has $h(x) = 0$ for $|x| > \lambda$. Let then

$$f(x) := \sum_{\Gamma_+ \cap \mathbb{Z}} \mu(\gamma)\, h(\gamma\, u)$$

where $\mu(n)$ is the Moebius function which equals $(-1)^k$ if n is a product of k primes and vanishes otherwise. By construction one has $f(x) = 0$ for $|x| > \lambda$, i.e. $f \in P_\lambda$, and

$$\sum_{\Gamma_+ \cap \mathbb{Z}} f(\gamma\, u) = h(u).$$

Thus by (4.10), one gets $\eta_S(f) = g$ which shows (ii).
(iii) This follows since 1_{R_S} is its own Fourier transform.

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

(iv) Follows from (ii) and (iii). □

We let $\mathcal{U}_S := \mathbb{F}_\mu \circ w_S$ and $\mathcal{M}_S : L^2(\mathbb{R}) \to L^2(\mathbb{R}, \mathrm{d}m_S)$ the unitary given by

$$\mathcal{M}_S(f)(s) := \left(\prod_{v \in S} L_v\left(\frac{1}{2} - is\right)\right)^{-1} f(s), \quad \mathrm{d}m_S(s) := \left|\prod_{v \in S} L_v\left(\frac{1}{2} - is\right)\right|^2 \mathrm{d}s \tag{4.14}$$

We let K_S be the kernel of the module $C_S \to \mathbb{R}_+^*$ and denote by $\mathbb{S}$ the generator of the scaling action of $\mathrm{GL}_1(\mathbb{R})_+ \subset \mathrm{GL}_1(\mathbb{A}_S) = \prod_{p \in S} \mathrm{GL}_1(\mathbb{Q}_p)$ on $L^2(X_S)^{K_S}$.

Proposition 4.2 (i) *The unitary transformation $\mathcal{V}_S := \mathcal{M}_S \circ \mathcal{U}_S : L^2(X_S)^{K_S} \to L^2(\mathbb{R}, \mathrm{d}m_S)$ gives the canonical form of the cyclic pair (D, ξ) where $D := \mathbb{S}$ and $\xi = \xi_S := \eta_S(\xi_\infty)$.*
(ii) *The cyclic pair $(\mathbb{S}, \xi_S)$ is even and the grading is given by the Fourier transform $\mathbb{F}_S$ which becomes the symmetry $s \mapsto -s$ under the unitary transformation $\mathcal{V}_S$.*

Proof (i) The unitary $\mathcal{U}_S : L^2(X_S)^{K_S} \to L^2(\mathbb{R})$ is an isomorphism and transforms the operator $\mathbb{S}$ into the multiplication by the variable s as in the proof of Proposition 3.1. By (4.13) applied to ξ_∞ one has

$$\begin{aligned}
\mathcal{U}_S(\xi_S)(s) &= \mathcal{U}_S(\eta_S(\xi_\infty))(s) = \left(\prod_{p \in S \setminus \{\infty\}} L_p\left(\frac{1}{2} - is\right)\right)(\mathbb{F}_\mu w_\infty(\xi_\infty))(s) \\
&= \prod_{v \in S} L_v\left(\frac{1}{2} - is\right) \\
\mathcal{V}_S(\xi_S)(s) &= \mathcal{M}_S \circ \mathcal{U}_S(h_S)(s) = 1.
\end{aligned}$$

Thus $\mathcal{V}_S : L^2(X_S)^{K_S} \to L^2(\mathbb{R}, \mathrm{d}m_S)$ is an isomorphism which transforms the operator $\mathbb{S}$ into the multiplication by the variable s and the vector ξ_S into the constant function 1.
(ii) One checks directly that the Fourier transform $\mathbb{F}_S$ anticommutes with $\mathbb{S}$. It fixes the vector ξ by Proposition 4.1. The uniqueness of the grading shows that the Fourier transform $\mathbb{F}_S$ becomes the symmetry $s \mapsto -s$ under the unitary transformation $\mathcal{V}_S$. □

We let $\iota_S : L^2(\mathbb{R}, \mathrm{d}m) \to L^2(\mathbb{R}, \mathrm{d}m_S)$ be the identity map $\iota_S(f)(s) := f(s), \forall s \in \mathbb{R}$.

Proposition 4.3 (i) *The map ι_S is bounded with bounded inverse.*
(ii) *One has the following commutative diagram:*

$$\begin{tikzcd}
L^2(\mathbb{R})^{ev} \arrow[r, "\eta_S"] \arrow[d, "\mathcal{V}"] & L^2(X_S)^{K_S} \arrow[d, "\mathcal{V}_S"] \\
L^2(\mathbb{R}, \mathrm{d}m) \arrow[r, "\iota_S"] & L^2(\mathbb{R}, \mathrm{d}m_S)
\end{tikzcd} \tag{4.15}$$

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

Proof (i) This follows since the function $\prod_{p\in S\setminus\{\infty\}} L_p(\frac{1}{2}-is)$ is bounded with bounded inverse, so that the Radon–Nikodym derivatives $\mathrm{d}m/\mathrm{d}m_S$ and $\mathrm{d}m_S/\mathrm{d}m$ are both bounded.
(ii) This follows from (4.13). $\square$

4.3 Semilocal Hermite operator

Let S be a finite collection of places (including the archimedean one) and $(\mathbb{S}, \xi_S)$ the cyclic pair of Proposition 4.2. As in Sect. 2, the associated Hermite operator N_S is defined as the grading operator associated with the filtration (E_n) of the Hilbert space $L^2(X_S)^{K_S}$ by the subspaces generated by the iterates $\mathbb{S}^j\xi$, $0 \le j \le n$.

Theorem 4.1 1. *For $S = \{\infty\}$ the semilocal Hermite operator N_S is the restriction of the harmonic oscillator to $L^2(\mathbb{R})^{ev}$.*
2. *The eigenfunctions of the semilocal Hermite operator N_S are elements of $L^2(X_S)^{K_S}$ of the form $\eta_S(P_n^S(x)e^{-\pi x^2})$, where P_n^S are polynomials obtained by orthonormalization and induction.*
3. *The matrix of $\mathbb{S}$ in the above orthonormal basis of $L^2(X_S)^{K_S}$, is a Jacobi hermitian matrix.*

Proof 1. See Proposition 3.2.
2. One has $\xi_S = 2^{1/4}\,\eta_S(e^{-\pi x^2})$ and the operator $\mathbb{S} = -i(H+\frac{1}{2})$ acts as scaling on the archimedean variable. The subspaces $E_n^S \subset L^2(X_S)^{K_S}$ are obtained by iteration of the operator $\mathbb{S}$ applied to the function $\eta_S(e^{-\pi x^2})$ and the scaling operator commutes with η_S. Thus E_n^S is the image by η_S of the space of products $P(x)e^{-\pi x^2}$ where P is an even polynomial of degree $\le 2n$. Since η_S is not unitary the orthogonalisation process delivers polynomials P_n^S which depend upon S.
3. Follows from (2.3), i.e. the general theory of orthogonal polynomials. $\square$

Thus we have reached a candidate for the analogue of the Hermite operator in the semilocal case and also for the prolate operator as in Definition 2.2:

$$\mathbf{W}_{\lambda,S} = \left(H+\frac{1}{2}\right)^2 + \lambda^2 N_S.$$

Remark 4.2 (i) Due to the non-unitary nature of η_S one has $N_S \circ \eta_S \neq \eta_S \circ N_\infty$ unless $S = \{\infty\}$ but the filtrations associated to the subspaces $N \le n$ correspond to each other by the map η_S as shown in Theorem 4.1 (ii).
(ii) Let $|\bullet|_S : X_S = \mathbb{A}_S/\Gamma \to \mathbb{R}_+$ be the module as in (4.5). Another guess for the semilocal analogue of the Hermite operator is to express the latter when $S = \{\infty\}$ in terms of a sum of the multiplication by $|x|_S^2$ and its conjugate under the Fourier transform.
(iii) Note that unless $S = \{\infty\}$ one has

$$|\bullet|_S^2\,\eta_S(f) \neq \eta_S(|\bullet|^2 f). \tag{4.16}$$

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

Indeed, taking $S = \{p, \infty\}$ for simplicity, one has for f an even function

$$w_S \eta_S(f) = \mathcal{E}_p(f), \quad \mathcal{E}_p(f)(u) = u^{1/2} \sum_{\mathbb{N}} f(p^n u)$$

the image of the l.h.s. of (4.16) under the map w_S is the function $u^2 \mathcal{E}_p(f)(u)$, while the image of the r.h.s. of (4.16) under the map w_S is $\mathcal{E}_p(|\bullet|^2 f)$ which evaluated at u gives the different expression

$$u^{1/2} \sum_{\mathbb{N}} u^2 p^{2n} f(p^n u)$$

4.4 The dual Hardy–Titchmarsh transform

We let as above S be a finite set of places containing ∞, and $\mathcal{U}_S := \mathbb{F}_\mu \circ w_S$. We let

$$E_S(s) := \prod_{p \in S} L_p\left(\frac{1}{2} + is\right) \tag{4.17}$$

and consider the Hilbert space $L^2\left(\mathbb{R}, \frac{ds}{|E_S(s)|^2}\right)$. We let $\beta_S : L^2(\mathbb{R}) \to L^2\left(\mathbb{R}, \frac{ds}{|E_S((s)|^2}\right)$ be the unitary given by

$$\beta_S(f)(s) := \left(\prod_{v \in S} L_v\left(\frac{1}{2} + is\right)\right) f(s) \tag{4.18}$$

Definition 4.3 The dual Hardy–Titchmarsh transform is the unitary

$$\upsilon_S : L^2(X_S)^{K_S} \to L^2\left(\mathbb{R}, \frac{ds}{|E_S(s)|^2}\right), \quad \upsilon_S = \beta_S \circ \mathcal{U}_S \tag{4.19}$$

Proposition 4.4 (i) *The following equality defines a sesquilinear pairing* $\langle L^2\left(\mathbb{R}, \frac{ds}{|E_S(s)|^2}\right) | L^2(\mathbb{R}, dm_S)\rangle$

$$\langle \xi | \eta \rangle_{\mathbb{R}} := \int \xi(x)\overline{\eta(x)} dx\,, \quad \forall \xi \in L^2\left(\mathbb{R}, \frac{ds}{|E_S(s)|^2}\right), \ \eta \in L^2(\mathbb{R}, dm_S) \tag{4.20}$$

(ii) *For any* $\xi, \eta \in L^2(X_S)^{K_S}$ *one has* $\langle \upsilon_S \xi | \mathcal{V}_S \eta \rangle_{\mathbb{R}} = \langle \xi | \eta \rangle$.

Proof The proof of (i) and (ii) follows from the equality valid for any place v and $s \in \mathbb{R}$,

$$\overline{L_v\left(\frac{1}{2} + is\right)} = L_v(\frac{1}{2} - is).$$

and combining (4.14) with (4.18). □

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

4.5 The Sonin space in the local case

The local definition of Sonin's space is as follows

Definition 4.4 Let $\mathbb{K}$ be a local field and α an additive character of $\mathbb{K}$. Let $\lambda > 0$. The Sonin space $\mathbf{S}_\lambda(\mathbb{K}, \alpha)$ is the subspace of the L^2-space of square integrable functions on $\mathbb{K}$ defined as follows

$$\mathbf{S}_\lambda(\mathbb{K}, \alpha) := \{f \in L^2(\mathbb{K}) \mid f(x) = 0 \text{ and } \mathbb{F}_\alpha f(x) = 0 \quad \forall x, |x| < \lambda\}$$

where $\mathbb{F}_\alpha$ denotes the Fourier transform with respect to α.

We use the imaginary exponential $e : \mathbb{Q}/\mathbb{Z} \to \{z \in \mathbb{C} \mid |z| = 1\}$, $e(x) := \exp(2\pi i x)$. We endow $\mathbb{Q}_p$ with the additive character (obtained using the embedding $\mathbb{Q}_p/\mathbb{Z}_p \subset \mathbb{Q}/\mathbb{Z}$)

$$e_p : \mathbb{Q}_p \to \mathbb{Q}_p/\mathbb{Z}_p \subset \mathbb{Q}/\mathbb{Z} \xrightarrow{e} \{z \in \mathbb{C} \mid |z| = 1\} \subset \mathbb{C} \tag{4.21}$$

which is equal to 1 on the maximal compact subring $\mathbb{Z}_p \subset \mathbb{Q}_p$.

Given a function $f \in L^2(\mathbb{K})$ and $a \in \mathbb{K}^*$, we let f_a be the function $f_a(x) := f(ax)$. The Fourier transform fulfills the equality

$$\mathbb{F}(f_a) = \frac{1}{|a|}\,\mathbb{F}(f)_{a^{-1}}$$

Proposition 4.5 *Let p be a finite prime and $\mathbb{K} = \mathbb{Q}_p$, $\alpha = e_p$. The $\mathbb{Z}_p^*$-invariant part of $\mathbf{S}_1(\mathbb{Q}_p, e_p)$ is one dimensional, with generator $\sigma_p := \epsilon_0 - \frac{1}{p}\epsilon_1$ where ϵ_n is the characteristic function of $\{x \in \mathbb{Q}_p \mid |x| = p^n\}$. Moreover one has $\mathbb{F}_{e_p}\sigma_p = \sigma_p$.*

Proof The function ϵ_n is equal to $(1_{\mathbb{Z}_p})_{p^n} - (1_{\mathbb{Z}_p})_{p^{n-1}}$ since one has

$$\epsilon_n(x) = 1_{\mathbb{Z}_p}(p^n x) - 1_{\mathbb{Z}_p}(p^{n-1}x).$$

Moreover one has $1_{\mathbb{Z}_p} = \sum_0^\infty \epsilon_{-\ell}$ and $(\epsilon_\ell)_{p^k} = \epsilon_{\ell+k}$. The Fourier transform of the function ϵ_n is then given by

$$\mathbb{F}_{e_p}(\epsilon_n) = p^n(1_{\mathbb{Z}_p})_{p^{-n}} - p^{n-1}(1_{\mathbb{Z}_p})_{p^{-n+1}} = p^n(1 - \frac{1}{p})\sum_0^\infty \epsilon_{-n-k} - p^n\frac{1}{p}\epsilon_{-n+1}. \tag{4.22}$$

By (4.22) one has

$$\begin{aligned}
\mathbb{F}_{e_p}\sigma_p &= \mathbb{F}_{e_p}\epsilon_0 - \frac{1}{p}\mathbb{F}_{e_p}\epsilon_1 \\
&= \left(\left(1 - \frac{1}{p}\right)\sum_0^\infty \epsilon_{-k} - \frac{1}{p}\epsilon_1\right) - \left(\left(1 - \frac{1}{p}\right)\sum_0^\infty \epsilon_{-1-k} - \frac{1}{p}\epsilon_0\right) \\
&= \epsilon_0 - \frac{1}{p}\epsilon_1
\end{aligned}$$

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

thus $\mathbb{F}_{e_p}\sigma_p = \sigma_p$. By construction one has $\sigma_p(x) = 0\ \forall x, |x| < 1$ thus $\sigma_p \in \mathbf{S}_1(\mathbb{Q}_p, e_p)$.

Let us now show that any $\psi \in \mathbf{S}_1(\mathbb{Q}_p, e_p)$ which is $\mathbb{Z}_p^*$-invariant is proportional to σ_p. Since ψ is $\mathbb{Z}_p^*$-invariant, there exists coefficients $a_n \in \mathbb{C}$ such that $\psi = \sum_{n\in\mathbb{Z}} a_n\epsilon_n$ and the L^2 condition means that $\sum p^n|a_n|^2 < \infty$. Since $\psi(x) = 0$ when $|x| < 1$ all the coefficients a_n for $n < 0$ vanish and one has $\psi = \sum_{n\geq 0} a_n\epsilon_n$. We now compute the Fourier transform $\mathbb{F}_{e_p}\psi$, which by (4.22) is

$$\mathbb{F}_{e_p}\psi = \sum_{n\geq 0} a_n \left(p^n\left(1-\frac{1}{p}\right)\sum_{k=0}^{\infty}\epsilon_{-n-k} - p^n\frac{1}{p}\epsilon_{-n+1}\right).$$

The sum on the right only involves ϵ_j for $j < 0$ except for the terms where $n = 0$ and $k = 0$ in the sum $\sum_0^\infty \epsilon_{-n-k}$ and the term with $n = 0$ and $n = 1$ in $-p^n\frac{1}{p}\epsilon_{-n+1}$. Thus the terms not involving ϵ_j for $j < 0$ are given by the expression $a_0(1-\frac{1}{p})\epsilon_0 - a_1\epsilon_0 - a_0\frac{1}{p}\epsilon_1$. Since $\psi \in \mathbf{S}_1(\mathbb{Q}_p, e_p)$ the sum of the terms involving the ϵ_j for $j < 0$ is equal to 0 and thus one gets the equality

$$\begin{aligned}\mathbb{F}_{e_p}\psi &= a_0\left(1-\frac{1}{p}\right)\epsilon_0 - a_1\epsilon_0 - a_0\frac{1}{p}\epsilon_1 = \left(a_0\left(1-\frac{1}{p}\right) - a_1\right)\epsilon_0 - a_0\frac{1}{p}\epsilon_1\\ &= a\epsilon_0 + b\epsilon_1.\end{aligned}$$

Finally in order that $\mathbb{F}_{e_p}(a\epsilon_0 + b\epsilon_1)$ vanishes for $|x| < 1$ requires that it vanishes for $x = 0$ i.e. that $\int(a\epsilon_0(x) + b\epsilon_1(x))\mathrm{d}x = 0$ which means that $a + pb = 0$. This gives $\left(a_0(1-\frac{1}{p}) - a_1\right) - a_0 = 0$, i.e. $a_0 = -pa_1$, so that ψ is a scalar multiple of σ_p. $\square$

4.6 The Sonin space in the semilocal case

Let S be as above, and $|\bullet|_S : X_S = \mathbb{A}_S/\Gamma \to \mathbb{R}_+$ be the module as in (4.5). We let α be the character of the additive group $\mathbb{A}_S = \prod_S \mathbb{Q}_v$ obtained as the product of the e_p (see (4.21)) with $e_\infty(x) := \exp(2\pi i x)$. We let $\sigma_S := \otimes_{S\setminus\{\infty\}}\sigma_p$.

Definition 4.5 Let $\lambda > 0$. The semilocal Sonin space $\mathbf{S}_\lambda(X_S, \alpha)$ is the subspace of the Hilbert space $L^2(X_S)^{K_S}$ defined as follows

$$\mathbf{S}_\lambda(X_S, \alpha) := \{f \in L^2(X_S)^{K_S} \mid f(x) = 0 \text{ and } \mathbb{F}_S f(x) = 0 \quad \forall x, |x| < \lambda\}$$

where $\mathbb{F}_S$ denotes the Fourier transform with respect to α.

Proposition 4.6 (i) *Let $f \in \mathbf{S}_\lambda(\mathbb{R}, e_\infty)$, $\theta_S(f)$ be the class of the function $\sigma_S \otimes f$ in $L^2(X_S)$. Then $\theta_S(f)$ belongs to $\mathbf{S}_\lambda(X_S, \alpha)$.*

(ii) *One has*

$$\mathbb{F}_\mu(w_S(\theta_S(f)))(s) = \mathbb{F}_\mu(w_\infty(f))(s) \times \prod_{S\setminus\{\infty\}}\left(1 - p^{-\frac{1}{2}-is}\right) \tag{4.23}$$

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

Proof (i) Let $z=(x,y)\in R_S\times\mathbb{R}=\mathbb{A}_S$ be such that $|z|=|x|_{S\setminus\{\infty\}}|y|_\infty<\lambda$. Then either $|x|_{S\setminus\{\infty\}}<1$ or $|y|_\infty<\lambda$. In both cases one gets $\sigma_S(x)f(y)=0$ as well as $\widehat{\sigma_S}(x)\widehat{f}(y)=0$. The same holds for the elements $\gamma z\in\mathbb{A}_S$, $\gamma\in\Gamma_+$ so that the class of $\sigma_S\otimes f$ in $L^2(X_S)$ belongs to $\mathbf{S}_\lambda(X_S,\alpha)$.
(ii) For simplicity of notation we assume $S=\{p,\infty\}$. One has, as in (4.9),

$$w_S(\sigma_S\otimes f)(1\times\lambda):=\lambda^{1/2}\sum_{n\in\mathbb{Z}}(\sigma_p\otimes f)(p^n,p^n\lambda)=\lambda^{1/2}\left(f(\lambda)-\frac{1}{p}f(\lambda/p)\right)$$

since $\sigma_p=\epsilon_0-\frac{1}{p}\epsilon_1$. Let $g=w_\infty(f)$ then one has

$$\lambda^{1/2}f(\lambda/p)=p^{1/2}(\lambda/p)^{1/2}f(\lambda/p)=p^{1/2}g(\lambda/p)$$

so that one gets

$$w_S(\sigma_S\otimes f)(1\times\lambda)=g(\lambda)-p^{-1/2}g(\lambda/p).$$

Next one has

$$\mathbb{F}_\mu(g)(s)=\int g(\lambda)\lambda^{-is}d^*\lambda$$

so that with $g_1(\lambda)=g(\lambda/p)$ one gets with $\lambda=pu$,

$$\begin{aligned}\mathbb{F}_\mu(g_1)(s)&=\int g_1(\lambda)\lambda^{-is}d^*\lambda=\int g(\lambda/p)\lambda^{-is}d^*\lambda\\&=\int g(u)(pu)^{-is}d^*u=p^{-is}\mathbb{F}_\mu(g)(s)\end{aligned}$$

which gives the required equality (4.23). □

4.7 The stability of Sonin spaces

We keep the above notation. Since $\left(1-p^{-\frac{1}{2}-is}\right)^{-1}=L_p(\frac{1}{2}+is)$, we rewrite (4.23) as

$$\mathbb{F}_\mu w_S(\theta_S(f))(s)=\left(\prod_{p\in S\setminus\{\infty\}}L_p\left(\frac{1}{2}+is\right)\right)^{-1}(\mathbb{F}_\mu w_\infty f)(s). \tag{4.24}$$

Proposition 4.7 *Let $\lambda>0$.*

(i) *Let $\mathbb{F}_S$ be the Fourier transform in $L^2(X_S)$. One has $\mathbb{F}_S\circ\theta_S=\theta_S\circ\mathbb{F}_{e_\mathbb{R}}$.*

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

(ii) *Let ι'_S be the map $f \mapsto f$, $L^2\left(\mathbb{R}, \frac{ds}{|E_\infty(s)|^2}\right) \to L^2\left(\mathbb{R}, \frac{ds}{|E_S(s)|^2}\right)$. One has the commutative diagram*

$$\begin{array}{ccc} L^2(\mathbb{R})^{ev} & \xrightarrow{\quad\theta_S\quad} & L^2(X_S)^{K_S} \\ \Big\downarrow{\scriptstyle \upsilon_\infty} & & \Big\downarrow{\scriptstyle \upsilon_S} \\ L^2\left(\mathbb{R}, \frac{ds}{|E_\infty(s)|^2}\right) & \xrightarrow{\quad\iota'_S\quad} & L^2\left(\mathbb{R}, \frac{ds}{|E_S(s)|^2}\right) \end{array} \tag{4.25}$$

(iii) *Let $f, g \in L^2(\mathbb{R})^{ev}$. One has $\langle \theta_S(f)|\eta_S(g)\rangle = \langle f|g\rangle$.*

Proof By Proposition 4.6, one has $\theta_S(\mathbf{S}_\lambda(\mathbb{R}, e_\infty)) \subset \mathbf{S}_\lambda(X_S, \alpha)$ where $\mathbf{S}_\lambda(X_S, \alpha)$ is the semilocal Sonin space.
(i) This follows from the equality $\mathbb{F}_{e_p}\sigma_p = \sigma_p$.
(ii) Let $f \in L^2(\mathbb{R})^{ev}$. We compute $\upsilon_S \circ \theta_S(f)$. By (4.19) one has $\upsilon_S = \beta_S \circ \mathfrak{U}_S = \beta_S \circ \mathbb{F}_\mu w_S$. By (4.24)

$$\begin{aligned} \upsilon_S \circ \theta_S(f)(s) &= \left(\beta_S \mathbb{F}_\mu w_S(\theta_S(f))\right)(s) \\ &= \left(\prod_{v \in S} L_v\left(\frac{1}{2} + is\right)\right)\left(\prod_{p \in S\setminus\{\infty\}} L_p\left(\frac{1}{2} + is\right)\right)^{-1} \\ (\mathbb{F}_\mu w_\infty f)(s) &= L_\infty\left(\frac{1}{2} + is\right)(\mathbb{F}_\mu w_\infty f)(s) = \upsilon_\infty(f)(s). \end{aligned}$$

(iii) The map $\mathfrak{U}_S = \mathbb{F}_\mu w_S : L^2(X_S)^{K_S} \to L^2(\mathbb{R})$ is unitary so that one has

$$\langle \theta_S(f)|\eta_S(g)\rangle = \langle \mathfrak{U}_S\theta_S(f)|\mathfrak{U}_S\eta_S(g)\rangle$$

and using (4.24) for the left term and (4.13) for the right term, one gets

$$\begin{aligned} &\langle \theta_S(f)|\eta_S(g)\rangle \\ &= \left\langle \left(\prod_{p \in S\setminus\{\infty\}} L_p\left(\frac{1}{2} + is\right)\right)^{-1} (\mathbb{F}_\mu w_\infty f) \Big| \left(\prod_{p \in S\setminus\{\infty\}} L_p\left(\frac{1}{2} - is\right)\right)(\mathbb{F}_\mu w_\infty g)\right\rangle \\ &= \langle f|g\rangle \end{aligned}$$

which gives the required equality. □

We are now ready to prove the following fact

Theorem 4.6 *Let $S \ni \infty$ be a finite set of places and $\lambda > 0$. Then the map θ_S is a hilbertian isomorphism of the Sonin spaces $\theta_S : \mathbf{S}_\lambda(\mathbb{R}, e_\infty) \to \mathbf{S}_\lambda(X_S, \alpha)$ where α is the normalized character.*

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

Proof We already know that $\theta_S(\mathbf{S}_\lambda(\mathbb{R}, e_\infty)) \subset \mathbf{S}_\lambda(X_S, \alpha)$ by Proposition 4.6 (i). To show that one has equality, let $h \in \mathbf{S}_\lambda(X_S, \alpha)$. The commutative diagram (4.25) shows that the map $\theta_S : L^2(\mathbb{R})^{ev} \to L^2(X_S)^{K_S}$ is bounded with bounded inverse, hence there exists a unique $f \in L^2(\mathbb{R})^{ev}$ such that $\theta_S(f) = h$. To show that $f \in \mathbf{S}_\lambda(\mathbb{R}, e_\infty)$ it is enough to show that f is orthogonal to all elements of P_λ and $\widehat{P_\lambda}$. By Proposition 4.1 (ii) and (iv), one has $\eta_S(P_\lambda) \subset P_\lambda^S$ and $\eta_S(\widehat{P_\lambda}) \subset \widehat{P_\lambda^S}$. Thus one obtains using Proposition 4.7 (iii) that for any element $g \in P_\lambda$, or $g \in \widehat{P_\lambda^S}$ one has

$$\langle f|g\rangle = \langle \theta_S(f)|\eta_S(g)\rangle = 0$$

since by construction the subspace $\mathbf{S}_\lambda(X_S, \alpha)$ is orthogonal to P_λ^S and to $\widehat{P_\lambda^S}$. Thus we conclude that $f \in \mathbf{S}_\lambda(\mathbb{R}, e_\infty)$. $\square$

4.8 Hilbertian spaces of entire functions

The stability exhibited by Theorem 4.6 shows that the hilbertian structure of the Sonin spaces $\mathbf{S}_\lambda(X_S, \alpha)$ is independent of S and the theory of Hilbert spaces of entire functions [12] in fact provides a canonical realization of these spaces which we now describe. It is important to point out that the choice of the finite set S plays a key role in fixing the inner product in the Hilbertian space.

4.8.1 Link with Hilbert spaces of entire functions

The link between Sonin spaces (called in [12] "Sonine spaces") and Hilbert spaces of entire functions, due to de Branges [11, 12], is that the map (cf. also Burnol's papers [2, 4])

$$\mathbf{S}_\lambda(\mathbb{R}, e_\infty) \ni f \mapsto \check{\mathcal{M}}(f)(z) := \pi^{-\frac{z}{2}}\Gamma\left(\frac{z}{2}\right)\int_{\mathbb{R}} f(t)t^{1-z}d^*t \tag{4.26}$$

sends $\mathbf{S}_\lambda(\mathbb{R}, e_\infty)$ onto a Hilbert space of entire functions, denoted here $\mathcal{B}_\lambda$, which belongs to the class of de Branges spaces.

4.8.2 Stability under semilocal amplification

The restriction of $\check{\mathcal{M}}(f)(z)$ to the critical line, i.e. $z = \frac{1}{2} + is$, coincides with the dual Hardy–Titchmarsh transform in the case $S = \{\infty\}$,

$$\check{\mathcal{M}}(f)\left(\frac{1}{2} + is\right) = \upsilon_\infty(f)(s) \tag{4.27}$$

as follows from Definition 4.3, (4.18), (4.19).

Proposition 4.8 *Let $S \ni \infty$ be a finite set of places and $\lambda > 0$. The map υ_S is an isomorphism of hilbertian spaces $\mathbf{S}_\lambda(X_S, \alpha)$ with $\mathcal{B}_\lambda$ and one has the commutative diagram*

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

$$\begin{tikzcd} \mathbf{S}_\lambda(\mathbb{R}, e_\infty) \arrow[rr, "\theta_S"] \arrow[dr, "\upsilon_\infty"'] & & \mathbf{S}_\lambda(X_S, \alpha) \arrow[dl, "\upsilon_S"] \\ & \mathcal{B}_\lambda & \end{tikzcd} \tag{4.28}$$

Proof This follows from the diagram (4.25) together with (4.27). $\square$

Note that $\mathcal{B}_\lambda$ inherits different inner products from its embedding in $L^2\left(\mathbb{R}, \frac{ds}{|E_S(s)|^2}\right)$.

4.8.3 De Branges space $\mathcal{B}_\lambda^S$

Let $z^\sharp$ designate the symmetric of z with respect to the critical line $L = \frac{1}{2} + i\mathbb{R}$. A de Branges space is a Hilbert space $\mathcal{K}$ of entire functions satisfying the following axioms:

(1) the evaluation at any $w \in \mathbb{C}$ is continuous;
(2) the map $F \mapsto F^\#$, $F^\#(w) = \overline{F\left(w^\#\right)}$ is a unitary anti-isometry of $\mathcal{K}$;
(3) if $F(\gamma) = 0$ then $G(w) = \left(w - \gamma^\#\right)/(w-\gamma)F(w)$ belongs to $\mathcal{K}$ and $\|G\| = \|F\|$.

It follows from these axioms that for any finite $S \ni \infty$ the space $\mathcal{B}_\lambda^S$ obtained by endowing the space $\mathcal{B}_\lambda$ with the norm induced by its embedding in $L^2\left(\mathbb{R}, \frac{ds}{|E_S(s)|^2}\right)$ is a de Branges space.

5 Metaplectic representation in the Jacobi picture

5.1 Infinitesimal representation and prolate operator

We recall that the metaplectic representation ϖ of $\mathfrak{sl}(2, \mathbb{R})$ on $L^2(\mathbb{R})$ is realized at the infinitesimal level by letting the standard basis $\{h, e_+, e_-\}$, with $[h, e_+] = 2e_+$, $[h, e_-] = -2e_-$, $[e_+, e_-] = h$, act on $\mathcal{S}(\mathbb{R})$, via the differential operators

$$\varpi(h) := x\partial_x + \frac{1}{2}, \quad \varpi(e_+) := i\pi x^2, \quad \varpi(e_-) := \frac{i}{4\pi}\partial_x^2. \tag{5.1}$$

Thus $\varpi(h)$ is up to a factor of i the scaling $\mathbb{S}$ while the Hermite operator, is obtained as $\varpi(k)$ using the generator $k = i(e_- - e_+)$ of the maximal compact subgroup $K = \widetilde{SO}(2)$. With these notations one has

$$\varpi(k) = \pi x^2 - \frac{1}{4\pi}\partial_x^2 \tag{5.2}$$

is the Hermite operator with spectrum $\{n + \frac{1}{2};\ n \in \mathbb{Z}^+\}$ and the Hermite functions

$$h_n = \left(2^{2n-\frac{1}{2}} n!\, \pi^n\right)^{-1/2} P_n(x) e^{-\pi x^2/2}, \quad \text{where} \quad P_n = (-1)^n e^{2\pi x^2} \partial_x^n \left(e^{-2\pi x^2}\right),$$

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

provide the associated eigenfunctions forming an orthonormal basis.

Thus, switching to the other standard $\mathbb{C}$-basis $\{k, n_+, n_-\}$, where $n_\pm = \frac{1}{2}\big(h \mp i(e_+ + e_-)\big)$, the representation ϖ is determined by (5.2) together with

$$\varpi(n_+) = \frac{\partial^2}{8\pi} + \frac{x\,\partial}{2} + \frac{1}{2}\pi x^2 + \frac{1}{4} = \left(\frac{\partial}{2\sqrt{2\pi}} + x\sqrt{\frac{\pi}{2}}\right)^2 = a^2 \tag{5.3}$$

$$\varpi(n_-) = -\frac{\partial^2}{8\pi} + \frac{x\,\partial}{2} - \frac{1}{2}\pi x^2 + \frac{1}{4} = -\left(-\frac{\partial}{2\sqrt{2\pi}} + x\sqrt{\frac{\pi}{2}}\right)^2 = -(a^*)^2 \tag{5.4}$$

where $a = \frac{\partial}{2\sqrt{2\pi}} + x\sqrt{\frac{\pi}{2}}$ and $a^* = -\frac{\partial}{2\sqrt{2\pi}} + x\sqrt{\frac{\pi}{2}}$ are the *annihilation* and *creation* operators, in terms of which $\varpi(k) = aa^* + a^*a$. They function as *lowering* and *raising* operators with respect to the basis $\{h_n;\ n \in \mathbb{Z}^+\}$ since

$$a(h_0) = 0, \quad a(h_n) = \sqrt{\frac{n}{2}}\,h_{n-1}, \quad \text{and} \quad a^*(h_n) = \sqrt{\frac{n+1}{2}}\,h_{n+1} \quad \forall n \in \mathbb{Z}^+. \tag{5.5}$$

The Casimir operator $C := h^2 + 2(e_+e_- + e_-e_+)$ fulfills $\varpi(C) = -\frac{3}{4}$. In particular this shows that $L^2(\mathbb{R}) = L^2(\mathbb{R})^{ev} \oplus L^2(\mathbb{R})^{odd}$ represents the decomposition of ϖ into two irreducible subrepresentations of lowest weight $\frac{1}{2}$ and $\frac{3}{2}$ respectively.

Proposition 5.1 *At the formal level the operator* $\mathbf{W}_\lambda$ *is of the form* $\varpi(\mathcal{W}_\lambda)$ *where* $\mathcal{W}_\lambda$ *is the following element of the enveloping algebra* $\mathfrak{U}(\mathfrak{sl}(2,\mathbb{R}))$

$$\mathcal{W}_\lambda = h^2 + 4\pi\lambda^2 k - \frac{1}{4}.$$

Proof This follows using (1.5), (5.1) and (5.2). □

5.2 Metaplectic representation and orthogonal polynomials

We shall now present the metaplectic representation as a special case of a general construction for orthogonal polynomials. We first consider, as in Sect. 2.2, the general set-up of orthogonal polynomials with respect to a measure $d\mu$ on $\mathbb{R}$ such that $d\mu(-s) = d\mu(s)$. Modulo a change of phase in the orthonormal basis we may assume that the coefficients a_n of the Jacobi matrix A for the multiplication by the variable $s \in \mathbb{R}$ are positive,

$$A = \begin{pmatrix} 0 & a_0 & 0 & 0 & 0 & \dots \\ a_0 & 0 & a_1 & 0 & 0 & \dots \\ 0 & a_1 & 0 & a_2 & 0 & \dots \\ 0 & 0 & a_2 & 0 & a_3 & \dots \\ 0 & 0 & 0 & a_3 & 0 & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots \end{pmatrix} \tag{5.6}$$

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

The coefficients a_n are determined by the moments $c_n := \int t^n d\mu(t)$ by (see [18, 2.2.7, 2.2.15])

$$a_n = \frac{k_n}{k_{n+1}}, \quad k_n = (D_{n-1}/D_n)^{1/2} \tag{5.7}$$

where the D_n are the determinants of Hankel matrices of the form, since the odd moments vanish,

$$D_5 = \begin{pmatrix} c_0 & 0 & c_2 & 0 & c_4 & 0 \\ 0 & c_2 & 0 & c_4 & 0 & c_6 \\ c_2 & 0 & c_4 & 0 & c_6 & 0 \\ 0 & c_4 & 0 & c_6 & 0 & c_8 \\ c_4 & 0 & c_6 & 0 & c_8 & 0 \\ 0 & c_6 & 0 & c_8 & 0 & c_{10} \end{pmatrix}$$

To investigate the Lie algebra generated by A and the grading operator N, $NP_n = nP_n$. We let

$$F_+ := N + \frac{1}{2i}[A, N], \quad F_- := N - \frac{1}{2i}[A, N] \tag{5.8}$$

Lemma 5.1 (i) *The double commutator* $[[A, N], N] = A$.
(ii) *The commutator* $[F_+, F_-]$ *is equal to* $-iA$.
(iii) *The double commutator* $[A, [A, N]]$ *is equal to* $f(N)$ *for the function*

$$f(n) = 2(a_{n-1}^2 - a_n^2) \tag{5.9}$$

(iv) *One has* $[A, F_+] = 2iF_+ + \Upsilon$ *where the diagonal matrix* Υ *has the entries* d_n *with*

$$\frac{1}{i} d_n = -2n + a_n^2 - a_{n-1}^2 \tag{5.10}$$

(v) *The matrix* $F_+ - \frac{1}{2}iA$ *is upper triangular and is the sum* $N - i\mathcal{S}$ *where* $\mathcal{S}$ *is the weighted shift*

$$\mathcal{S} = \begin{pmatrix} 0 & a_0 & 0 & 0 & 0 & 0 \\ 0 & 0 & a_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & a_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & a_3 & 0 \\ 0 & 0 & 0 & 0 & 0 & a_4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Proof (i) The commutator $[A, N]$ is of the form

$$\begin{pmatrix} 0 & a_0 & 0 & 0 & 0 & 0 \\ -a_0 & 0 & a_1 & 0 & 0 & 0 \\ 0 & -a_1 & 0 & a_2 & 0 & 0 \\ 0 & 0 & -a_2 & 0 & a_3 & 0 \\ 0 & 0 & 0 & -a_3 & 0 & a_4 \\ 0 & 0 & 0 & 0 & -a_4 & 0 \end{pmatrix} \tag{5.11}$$

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

and the double commutator $[[A, N], N] = A$.
(ii) One has

$$[F_+, F_-] = [N + \frac{1}{2i}[A, N], N - \frac{1}{2i}[A, N]] = \frac{1}{i}[[A, N], N] = -iA.$$

(iii) This important fact comes from computing $[A, [A, N]]$ which gives this matrix

$$\begin{pmatrix} -2a_0^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2a_0^2 - 2a_1^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2a_1^2 - 2a_2^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2a_2^2 - 2a_3^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2a_3^2 - 2a_4^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & \dots \end{pmatrix}$$

(iv) One has

$$[A, F_+] = [A, N] + \frac{1}{2i}[A, [A, N]] = 2iF_+ + \Upsilon, \quad d_n = -2in + \frac{1}{2i}f(n).$$

(v) By (5.11), the matrix $F_+ - \frac{1}{2}iA$ is upper triangular. □

We look for a representation σ of $\mathfrak{sl}(2, \mathbb{R})$ such that, with A as in (5.6) and $k = i(e_- - e_+)$,

$$\sigma(h) = -iA, \quad \sigma(k) = N + c \tag{5.12}$$

for some constant c. The identity

$$[h, [h, k]] = -2i([h, e_+] + [h, e_-]) = -4i(e_+ - e_-) = 4k \tag{5.13}$$

translates into the relation

$$[A, [A, \sigma(k)]] = -4\sigma(k). \tag{5.14}$$

By lemma 5.1 (iii), $[A, [A, N]]$ is the diagonal matrix with diagonal entries $2\big(a_{n-1}^2 - a_n^2\big)$. The identity (5.14) is thus equivalent to

$$2(a_{n-1}^2 - a_n^2) = -4(n + c), \quad \forall n \geq 0. \tag{5.15}$$

One has $[h, k] = -2i(e_+ + e_-)$ so that

$$e_+ = \frac{i}{2}\left(k + \frac{1}{2}[h, k]\right), \quad e_- = \frac{i}{2}\left(-k + \frac{1}{2}[h, k]\right) \tag{5.16}$$

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

Thus with the notation (5.8) one gets, using (5.12)

$$\sigma(e_+) = \frac{i}{2}(F_+ + c), \quad \sigma(e_-) = \frac{i}{2}(-F_- - c)$$

Thus by Lemma 5.1 (iv) one obtains

$$[\sigma(h), \sigma(e_+)] = \frac{1}{2}[A, F_+] = iF_+ + \frac{1}{2}\Upsilon.$$

The Lie algebra equality $[h, e_+] = 2e_+$ together with (5.10) and (5.15) then determine uniquely $c = \frac{1}{4}$. This uniquely determines the a_n, the representation σ, and $E_\pm := -i\sigma(e_\pm)$ are given by

$$E_+ = \frac{1}{2}\left(N + \frac{1}{4}\right) + \frac{1}{4}[A, N], \qquad E_- = -\frac{1}{2}\left(N + \frac{1}{4}\right) + \frac{1}{4}[A, N], \tag{5.17}$$

Theorem 5.2 (i) *The positive coefficients a_n are specified uniquely in order for* (5.12) *to define a representation of the Lie algebra $\mathfrak{sl}(2, \mathbb{R})$, one has*

$$a_n = \frac{1}{2}\sqrt{(2n+1)(2n+2)}. \tag{5.18}$$

(ii) *The Hamburger moment problem for the moments associated to the sequence a_n by* (5.7) *is determinate.*

(iii) *The unique measure $d\mu$ of the moment problem of (ii) is the probability measure proportional to $|\Gamma(\frac{1}{4} + \frac{1}{2}is)|^2 \mathrm{d}s$.*

(iv) *The representation σ is the even component of the metaplectic representation ϖ of $\widetilde{SL}(2, \mathbb{R})$.*

Proof (i) Using (5.15) together with $c = \frac{1}{4}$ one gets

$$2(a_{n-1}^2 - a_n^2) = -4n - 1$$

so that one obtains (5.18).

(ii) By [17, Thm. 2, p. 86] *the Hamburger moment problem* corresponding is determinate, as follows from [16, Corollary 6.19] since

$$\sum_{n=0}^{\infty} \frac{1}{a_n} \equiv \sum_{n=0}^{\infty} \frac{2}{\sqrt{(2n+1)(2n+2)}} = \infty.$$

(iii) The coefficients a_n agree with those given by Proposition 3.3, after rescaling by powers of i.

(iv) The eigenvalues of $\sigma(k)$, which serve as weights for the representation σ relative to the Cartan subgroup $\widetilde{S0}(2) \subset \widetilde{SL}(2, \mathbb{R})$, exactly reproduce the spectrum of the Hermite operator transported from $L^2(\mathbb{R})^{ev}$. This identifies the irreducible representation σ as being the even component of the metaplectic representation ϖ of $\widetilde{SL}(2, \mathbb{R})$. ◻

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

We now determine explicitly the moments $c_n := \int t^n d\mu(t)$ where $d\mu$ is the probability measure proportional to $|\Gamma(\frac{1}{4}+\frac{1}{2}is)|^2$. We use the equalities (5.7),

$$k_n = (D_{n-1}/D_n)^{1/2},\ \ a_n = \frac{k_n}{k_{n+1}},\ \ a_n^2 = \frac{1}{4}(2n+1)(2n+2),$$

which imply

$$D_{n-1}/D_n = k_n^2,\ \ k_n^2 = k_0^2 \prod_0^{n-1} a(j)^{-2} = \frac{4^n}{\Gamma(2n+1)}$$

This then determines the moments and shows that they are rational numbers, the first are

$$c_0 = 1,\ c_2 = \frac{1}{2},\ c_4 = \frac{7}{4},\ c_6 = \frac{139}{8},\ c_8 = \frac{5473}{16},\ c_{10} = \frac{357721}{32}$$

What one finds in general is that the denominators of the moments are 2^n for c_{2n} so that they are determined by a sequence of integers. The first ones are

$$\begin{aligned} &1,\ 7,\ 139,\ 5473,\ 357721,\ 34988647,\\ &\quad 4784061619,\ 871335013633,\ 203906055033841,\\ &\quad 59618325600871687,\ 21297483077038703899,\ 9127322584507530151393,\ldots \end{aligned}$$

This is a classical integer sequence, known as $A126156$ [19]. In fact one has

Proposition 5.2 *The normalized moments c_n of the measure $(2\pi)^{-3/2}|\Gamma(\frac{1}{4}+\frac{1}{2}is)|^2 \mathrm{d}s$ are such that $c_n = 0$ for n odd and*

$$\sum c_n \frac{(ix)^n}{n!} = \sqrt{\frac{2}{e^x + e^{-x}}}. \tag{5.19}$$

Proof The normalization means $c_0 = 1$ and that the measure $d\mu$ proportional to $|\Gamma(\frac{1}{4}+\frac{1}{2}is)|^2 \mathrm{d}s$ is a probability measure. The left hand side of (5.19) is then equal to

$$\sum c_n \frac{(ix)^n}{n!} = \int_{-\infty}^{\infty} \exp(isx) d\mu(s) = \langle \exp(ixh)\mathbb{F}_\mu(w(\xi)) | \mathbb{F}_\mu(w(\xi))\rangle$$

where $\xi \in L^2(\mathbb{R})^{ev}$, $\xi(x) = e^{-\pi x^2}$, and h is the self-adjoint operator of multiplication by s.

$$\mathbb{F}_\mu(w(\xi))(s) = \int_0^\infty x^{1/2-is} e^{-\pi x^2} \mathrm{d}^* x = \frac{1}{2}\pi^{-\frac{1}{4}+i\frac{s}{2}} \Gamma\left(\frac{1}{4} - i\frac{s}{2}\right)$$

Journal: 43034 Article No.: 0388 TYPESET DISK LE CP Disp.:2024/9/12 Pages: 38 Layout: Small-Ex

One has $(\mathbb{F}_\mu f_\lambda)(s) = \lambda^{is}(\mathbb{F}_\mu f)(s)$ where $f_\lambda(u) := f(\lambda u)$, and taking $\lambda = e^x$ one obtains

$$\sum c_n \frac{(ix)^n}{n!} = \langle \lambda^{1/2}\xi_\lambda | \xi \rangle = \lambda^{1/2} \int_{-\infty}^{\infty} e^{-\pi\lambda^2 t^2} e^{-\pi t^2} \mathrm{d}t = \lambda^{1/2}(1+\lambda^2)^{-1/2}$$

which gives the required equality. □

6 Appendix: Fourier transform on *p*-adic numbers

In Weil's book [22], one finds the following general definitions and facts about non-discrete locally compact fields.

Definition 6.1 A non-discrete locally compact field K will be called a p-field if p is a prime and $|p \cdot 1_K|_K < 1$, and an $\mathbb{R}$-field if it is an algebra over $\mathbb{R}$.

Definition 6.2 Let K be a p-field, R its maximal compact subring and P the maximal ideal of R. Then the order of a non-trivial character χ of K is the largest integer $v \in \mathbb{Z}$ such that χ is 1 on $P^{-\nu}$; it will be denoted by $\mathrm{ord}(\chi)$.

Corollary 6.3 [22] *Let R be the maximal compact subring of K, and φ the characteristic function of R. Let χ be a non-trivial character of K, of order ν, and let α be the self-dual Haar measure on K for the identification of K with its dual, based on χ. Let $a \in K^\times$ be such that $\mathrm{ord}_K(a) = \nu$. Then $\alpha(R) = \mathrm{mod}_K(a)^{1/2}$, and the Fourier transform of φ is $y \to \mathrm{mod}_K(a)^{1/2}\varphi(ay)$.*

On p-adic numbers $\mathbb{Q}_p$ the basic character χ_p is defined by $\chi_p(x) = \exp\left(2i\pi\{x\}_p\right)$ where $\{x\}_p$ is the fractional p-adic part of x. In particular it has order 0 in the sense of Definition 6.2. Thus by Corollary 6.3 the Fourier transform of φ, the characteristic function of R, is equal to φ and the self-dual Haar measure of the subset $R \subset \mathbb{Q}_p$ is equal to 1.

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Journal: **43034** Article No.: **0388** ☐ TYPESET ☐ DISK ☐ LE ☐ CP Disp.:**2024/9/12** Pages: **38** Layout: **Small-Ex**

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